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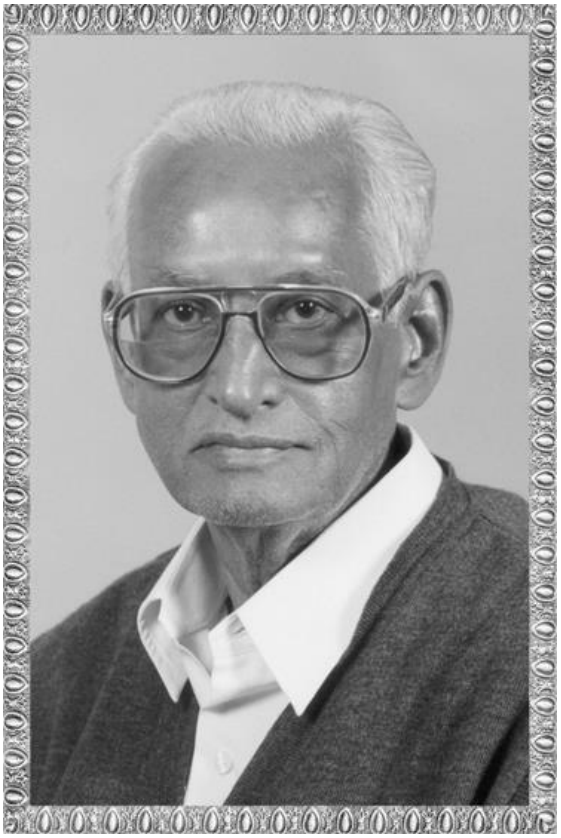

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BCA/BSCIT SEM-6 Cloud Computing

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Unit-1
Part1: OverviewofComputingParadigm

Whatiscomputing?

- Computingistheutilizationofcomputersto completeatask. Itinvolvesbothhardware& software functions performing some sort of task with a computer.
- Thisactivityincludesbothdevelopingthesoftwareandhardwareas well.
- Examplesofcomputingbeingusedineverydaylife:sendinganemail,swipingcredit/debit cards etc.
- Theutilizationofcomputertechnologyto completethetaskisknownas computing.

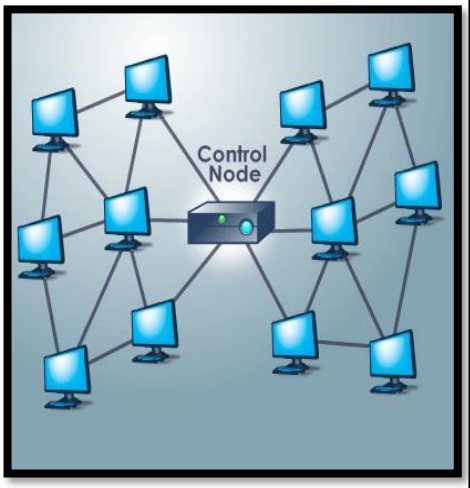
RecenttrendsinComputing

1. GridComputing

- Grid computing consists of a large number of computers which are connected parallel and forms a computer cluster.
- "theGrid"linkstogethercomputingresourceslikePCs,workstations,servers,storage elementsand provides themechanism needed to access them.
- Thiscombinationofconnectedcomputersusestosolveacomplex problem.
- Ingridcomputing,thisserviceofthecomputerisconnectedandrunindependenttasks. Grid computing uses a wide range of applications, for instance, mathematical, scientific, and educational tasks through several computing resources.
- ByimplementingourproposedIntranetGriditisveryeasytodownloadmultiplefiles very fast.
- No need to worry about the security as authenticatingis done each and every step taking place in Grid.

Advantagesofgridcomputing:-

- It can solve more complex problems in a very short span of time.
- Itcaneasilycombinewithotherorganization.
- Itcan makebetter useofexisting hardware.



Disadvantagesofgridcomputing:-

- Itevolves gridsoftwareand standards.
- Itstarts**learningcurve.**(The*learningcurve*showsthatif apersonperformssimilartaskagainandagain,thenafter a period of time there will be an improvement in his/her performance.)

- Itisvery noninteractive.

2. ClusterComputing

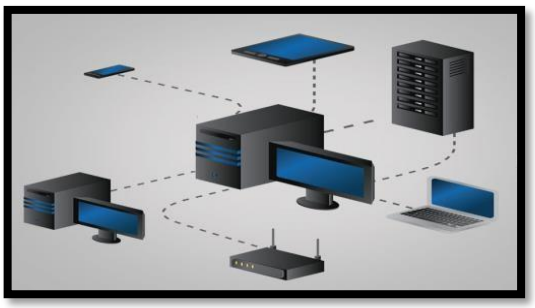
- ClustercomputingorHigh-Performancecomputingframeworksisaformofcomputing in which bunch of computers (nodes) that are connected through a LANso that, they behave like a single machine.
- Acomputerclusterhelptosolvecomplexoperationsmoreefficientlywithmuchfaster processing speed, better data integrity than a single computer and they only used for mission-critical applications.
- Some of the critical Applications of **Cluster Computers are Google Search Engine and Weather Forecasting.**
- Clustercanbeclassifiedintotwocategory**OpenandCloseCluster**.
- Open Cluster: **All nodes in Open Cluster are needed IPs**, and that are accessible through internet/web that cause more security concern.
- CloseCluster:OntheotherhandCloseClusterare**hidebehindthegatewaynode**and provide better security.

TypesofClustercomputing

1. **Load-balancing clusters:** As the name implies, This system is used to distribute workload across multiple computers. That system distributes the processing load as possible across a cluster of computers.
2. **High availability (HA) clusters:** A high availability clusters (HA cluster) are the bunch of computers that can reliably utilize for redundant operations in the event of nodes failure in Cluster computing.
3. **High performance (HP) clusters:** This computer networking methodology use supercomputers and Cluster computing to solve advanced computation problems.

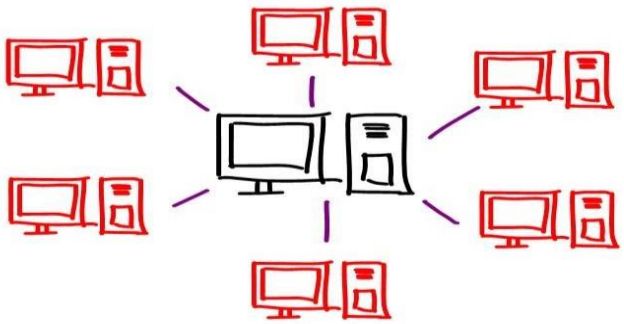
AdvantagesofusingClustercomputing

1. **Cost efficiency**: In a Cluster computing Cost efficiency is the ratio of cost to output, that is the connecting group of the computer as computer cluster much cheaper as compared to mainframe computers.
2. **Processing speed:** The Processing speed of computer cluster is the same as a mainframe computer.
3. **Expandability:**ThebestbenefitofClusterComputingisthatitcanbeexpandedeasily by adding the additional desktop workstation to the system.
4. **High availability of resources:** If any node fails in a computer cluster, anothernode within the cluster continue to provide uninterrupted processing.Whenamainframesystemfails,the entire system fails.



3. DistributedComputing

- ADistributedcomputingisamodelofcomputationthatistightlyrelatedtoDistributed Systems, refers to as multiple computer systems located at different places linked together over a network and use to solve higher level computation without having to use an expensive supercomputer.
- AllthecomputersaretiedtogetherinanetworkeitheraLocalAreaNetwork(LAN)or WideAreaNetwork(WAN),communicatingwitheachothersothatdifferentportions of a Distributed applications run on different computers from any geographical location.
- EverynodeontheDistributedcomputingis**autonomousmachines**(donotphysically share memory or processors but thereby sharing resources such as printers and databases).
- Distributed Systems have broken down into two parts: **the front end and the back end**.
- Grid computing is a based on distributed architecture and is the form of "distributed computing" or "peer-to-peer computing "that involving large numbers of computers physically connected to solve a complex problem.
- Standaloneapplicationsaretraditional applications(or3-tieroldsystems)thatrunona single system; distributed applications run on multiple systems simultaneously.
- Traditional applications need to be installed on every system and make it hard to maintain.However,InDistributedcomputing,applicationsrunonbothsimultaneously.
- With distributed computing, if a workstation that goes down, another workstation can resume the jobs.
- The **advantages of distributed computing** increased the speed with "absolute performance" and lower cost with more reliability than a non-distributed system.
- Itiscurrentlyquitepopular,andmanybusinessesareconvertingto it.
- Extremely well-known example of distributed systems and applications of distributed computingusedin**Telephonenetworksandcellularnetworks,Computernetworks such as the Internet, Wireless sensor networks, Routing algorithms.**



4. UtilityComputing

- The term **Utility Computing** refers to utility computing technologies and business models,whichprovidesaserviceprovidertoitscustomersITservices,andtheycharge you by consumption.
- ExamplesofsuchITservicesarecomputingpower,storageor applications.
- The term utility refers to utility services such as electricity, telephone, water and gas that are provided by a utility company.
- Similartotheelectricityortelephoneifthecustomerreceivestheutilitycomputing,the computing power on a shared computer network, its consumption is measured and billed on that basis.
- Utilitycomputingisverysimilartovirtualizationsothat thetotalamountofwebstorage space along with the computing power that is available to the users is much larger compared to that of the single time-sharing computer.
- Multiplebackendwebserversareoftenusedtomakethiskindofwebservicepossible.
- Thededicatedwebserversmaybeusedinclusterformswhicharespecificallycreated and then leased to the end users.
- The method of using a single such 'calculation' on multiple web servers is called the distributed computing.

AdvantagesofUtilityComputing

- UtilitycomputingreducesthecostofIT,giventhatexistingresourcescanbeusedmore effectively.
- The costs are transparent and the various departments of a company can be directlyassigned.
- Thecompaniesachievegreaterflexibility,becausetheirITresourcesmorequicklyand easily adapt to changeable demand.
- Overall, it is easier to manage the entire IT structure, as there will no longer be made for each application, which is a benefit for specific IT infrastructure.

5. CloudComputing

- Cloudcomputingisthe deliveryofdifferent servicesthroughtheInternet.
- These resources include tools and applications like data storage, servers, databases, networking, and software.
- Rather than keeping files on a proprietary hard drive or local storage device, cloud-based storage makes it possible to save them to a remote database.
- As long as an electronic device has access to the web, it has access to the data and the software programs to run it.
- Cloudcomputingisapopularoptionforpeopleandbusinessesforanumberofreasons includingcostsavings,increasedproductivity,speedandefficiency,performance,and security.
- Cloudcomputingcan beboth publicand private.

- Publiccloudservicesprovidetheirservicesoverthe Internetfora fee.
- Private cloud services, on the other hand, only provide services to a certain number of people. These services are a system of networks that supply hosted services.
- Thereis also a hybrid option, which combines elements of both the public and private services.
- Cloud computing is new service but is being used by a number of different organizationsfrom **bigcorporationstosmallbusinesses**,**nonprofitstogovernment agencies**, and even **individual consumers**.

TypesofCloudServices

- Regardlessofthekindofservice,cloudcomputingservicesprovideuserswithaseries of functions including:
  - Email
  - Storage,backup,anddata retrieval
  - Creatingandtestingapps
  - Analyzingdata
  - Audioand video streaming
  - Deliveringsoftwareondemand

Evolutionofcloudcomputing

- The trend toward cloud computing started **in the late 1980s** with the concept of grid computing when, forthe first time,alargenumber ofsystems were applied to asingle problem,usuallyscientificinnatureandrequiringexceptionallyhighlevelsofparallel computation.
- In Europe, long distance optical networks are used to tie multiple universities into a massive computing grid in order that resources could be shared and scaled for large scientific calculations.
- Gridcomputingprovideda**virtualpool**ofcomputationresourcesbutit'sdifferentthan cloud computing.
- Grid computing specifically used to take maximum advantages from connected computers in parallel to solve a particular, individual problem, or to run a specific application.
- Cloudcomputing, ontheotherhand, usedto take maximumadvantages frommultiple resources,includingcomputingresources,todeliver anintegrated"service"totheend user.
- **In grid computing**, the focus is on moving a workload to the location of the needed computing resources, which are mostly remote and are readily available for use. Usuallyagridisaclusterofserversonwhichalargetaskcouldbedividedintosmaller taskstoruninparallel.Fromthispointofview,agridcouldactuallybeviewedas just one virtual server. Grids also require applications to conform to the grid software interfaces.

- **In a cloud environment**, computing and extended IT and business resources, such as servers, storage, network, applications and processes, can be dynamically shaped or carved out from the underlying hardware infrastructure and made available to a workload.In addition,whileacloudcanprovisionandsupportagrid,acloudcanalso supportnon-gridenvironments,suchasathree-tier Webarchitecturerunning traditional or Web 2.0 applications
- **In the 1990s**, the concept of virtualization was expanded beyond virtual servers to higher levels of abstraction—first the virtual platform, including storage and network resources, and subsequently the virtual application, which has no specific underlying infrastructure. **Utility computing** offered clusters as virtual platforms for computing with a metered business model.
- More recently software as a service **(SaaS)** has raised the level of virtualization to the application, with a business model of charging not by the resources consumed but by the value of the application to subscribers. **The concept of cloud computing has evolvedfromtheconceptsofgrid,utilityandSaaS**.Itisanemergingmodelthrough which users can gain access to their applications from anywhere, at any time, through their connected devices.
- Companies can choose to share resources using **public or private clouds**, depending on their specific needs. **Public clouds** expose services to customers, businesses and consumers on the Internet. **Private clouds** are generally restricted to use within a company behind a firewall and have fewer security exposures as a result.
- Asmoreenterprisesaddcloudcomputingthelevelofapplicationsismigratingtoward more mission critical and SaaS will become a support of IT strategies.
- Anumberofcompanies,including**Google,Microsoft,Amazon,andIBM,**havebuilt enormousdatacenter-basedcomputingcapacityallovertheworldtosupporttheirWeb service offerings search, instant messaging, web-based retail.

Businessdriverforadoptingcloudcomputing

1. **Business growth**
   Business growth is one of the top benefits organizations realize as a result of cloud adoption,with52%ofenterprisesreportingincreasedgrowthsincegoingcloud(2015 Cloud EnterpriseReport).
2. **Efficiency**
   Efficiencyisanextremelycommonclouddriver,with71%oforganizationsworldwide ranking it a top area they hope to approve through cloud technology (2015 Cloud EnterpriseReport).
3. **Experience**
   Next among the business drivers is improving the quality of the customer experience, which45%ofenterprisesworldwiderankasatopclouddriver(althoughthatnumber jumps to 61%looking at only organizations in the UK and Australia).

4. **Assurance**
   Finally, there is assurance, which is the idea thatdata will be more securein the cloud andtheuserwillattainbetteruptimebecauseitssolutionsaremaintainedbyproviders that havebuilt their businesses around these competencies.
5. **Agility**
   Improving ITagility,orenablingITtobemoreresponsivetobusinessneedsandreact fastertomarketchanges,isatoptechnologydriverfor66%oforganizationsworldwide (2015 Cloud Enterprise Report). This goal is very possible in the cloud environment, as SaaS technologies mean that IT no longer needs to be consumed with traditional application management tasks.
6. **Cost**
   The cost driver has two sides: Reducing IT expenses and restructuring these expenses tospreadthemoutovertimethankstothelicensingmodelofSaaStechnologies.While 39%oforganizationsdoconsiderreducingITexpensesakeyclouddriver(2015Cloud Enterprise Report).

Clustercomputingv/sGridComputingv/sCloud computing

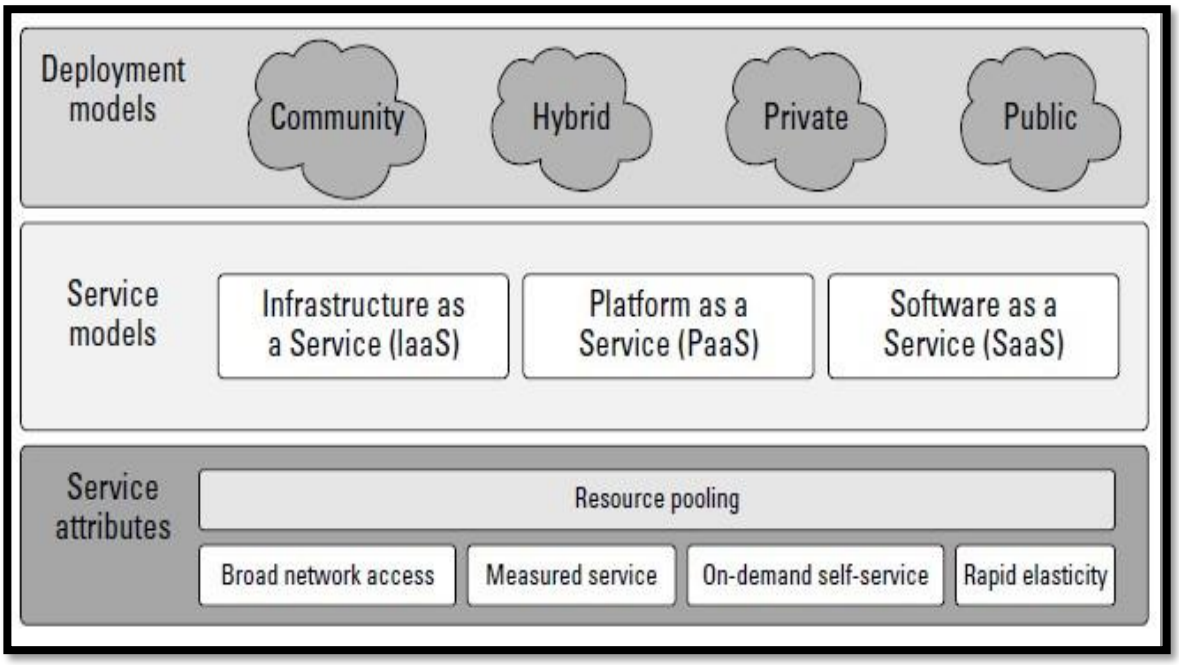
	ClusterComputing	**Grid Computing**	**Cloud Computing**
BasicIdea	Aggregation of resources.	Segregation of Resources.	Consolidation of Resources.
RunningProcesses	Same processes run onallcomputersover the cluster at the same time.	Jobisdividedintosub-jobs each is assigned toanidleCPUsothey all run concurrently.	Dependsonservice provisioning. Which computer offersaserviceand provisionsittothe requestingclients.
OperatingSystem	Allnodesmustrun thesameoperating system.	No restriction is made on the operating system.	No restriction is made on the operatingsystem.
Job Execution	Execution depends on job scheduling. So,jobswaitunitit's assigned a runtime.	Execution is scalable in a way that moves the execution of a job toanidleprocessor (node).	Self-Managed.
SuitableforApps	Cascadingtasks.If onetasksdependson anotherone.	Not suitable for cascading tasks.	On-demandservice provisioning.
Locationof nodes	Physically in the same location	Distributed geographicallyallover the globe.	Location doesn't matter

Homo/Heterogeneity	Homogenous	Heterogeneous	Heterogeneous
Virtualization	None	None	Virtualizationisa key
Transparency	Yes	Yes	Yes
Security	High	High, but doesn't reach the level of cluster computing.	Lowerthanboth types.
Interoperability	Yes	Yes	No
Application Domains	industrialsector, research centers, health care, and centers that offer servicesonthe nation-widelevel	industrial sector, research centers, health care, and centers that offer servicesonthenation-widelevel	Banking, Insurance,Weather Forecasting,Space Exploration, Business, IaaS, PaaS,SaaS
Implementation	Easy	Difficult	Difficult – needto be done by the host.
Management	Easy	Difficult	Difficult
Resource Management	Centralized(locally)	Distributed	Both centralized and distributed.
Internet	No internet access is required	Required	Required

Unit-1
Part2:IntroductiontoCloudComputing

CloudComputing(NISTModel)

- Cloudcomputingisamodelforenablingeverywhere,convenient,on-demandnetwork access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction.
- This cloud model is composed of five essential characteristics, three service models, and four deployment models



Serviceattributes:

- **On-demand self-service:** A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider.
- **Broad network access:** Capabilities are available over the network and accessed throughstandardmechanismsthatpromoteusebyheterogeneousthinorthickclient platforms (e.g., mobile phones, tablets, laptops, and workstations).
- **Resourcepooling:**Theprovider'scomputingresourcesarepooledtoservemultiple consumersusingamulti-tenantmodel,withdifferentphysicalandvirtualresources dynamicallyassignedandreassignedaccording toconsumer demand.Examples of resources include storage, processing, memory, and network bandwidth.

- **Rapid elasticity:** Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be appropriated in any quantity at any time.
- **Measuredservice**:Cloudsystemsautomaticallycontrolandoptimizeresourceuse by leveraging a metering capability1at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resourceusagecanbemonitored,controlled,andreported,providingtransparency for both the provider and consumer of the utilized service.

ServiceModels:

Cloudcomputingisbasedonservicemodels.Thesearecategorizedintothreebasicservice models which are -

- Infrastructure-as–a-Service(IaaS)
- Platform-as-a-Service(PaaS)
- Software-as-a-Service(SaaS)

Anything-as-a-Service (XaaS) is yet another service model, which includes Network-as-a-Service, Business-as-a-Service, Identity-as-a-Service, Database-as-a-ServiceorStrategy-as-a-Service.The**Infrastructure-as-a-Service (IaaS)**is the most basic level of service.

Infrastructure-as-a-Service(IaaS)

IaaSprovides access to fundamental resources such as physical machines, virtual machines, virtual storage, etc.

Platform-as-a-Service(PaaS)

PaaS provides the runtime environment for applications, development and deployment tools, etc.

Software-as-a-Service(SaaS)

SaaSmodelallowsto usesoftwareapplicationsas aservicetoend-users.

DeploymentModels:

PublicCloud
The**publiccloud**allowssystemsandservicestobeeasilyaccessibletothegeneralpublic. Public cloud may be less secure because of its openness.

PrivateCloud
The**private cloud**allows systems and services to be accessible within an organization. It is more secured because of its private nature.

CommunityCloud
The**communitycloud** allowssystemsandservicestobeaccessiblebyagroupof organizations.

HybridCloud

The**hybrid cloud** is a mixture of public and private cloud, in which the critical activities are performed using private cloud while the non-critical activities are performed using public cloud.

Historyofcloudcomputing

- TheconceptofCloudComputingcameintoexistencein1950withimplementationof mainframe computers, accessible via thin/static clients.
- Sincethen,cloudcomputinghasbeenevolvedfromstaticclientstodynamiconesfrom softwaretoservices.Thefollowingdiagramexplainstheevolutionofcloudcomputing:

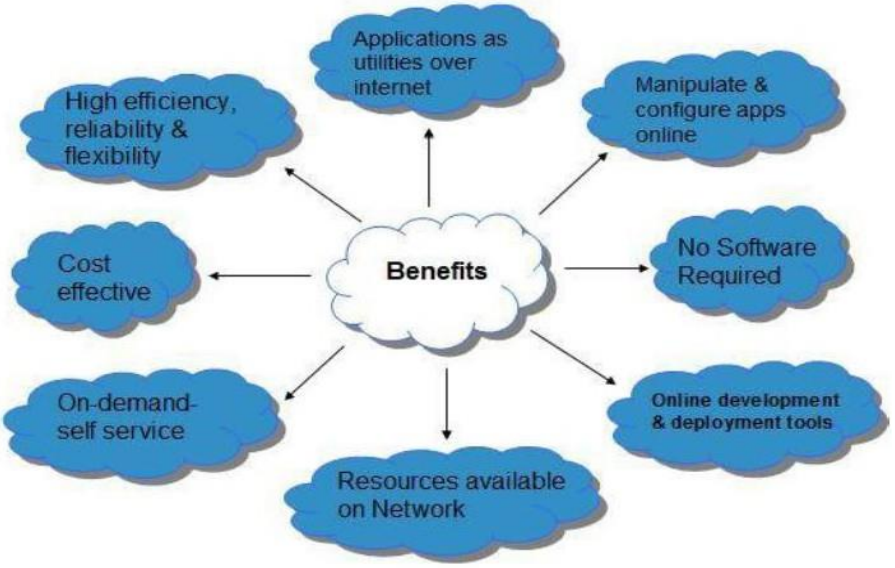
Date	Development
1950s	Large-scale servers at universities and companies were accessed by considerably weak terminals/clients.
1960s	John McCarthy foresaw that computing technology will be organized in the future like public electric and water utilities in 1960s.
1970s	Time sharing concept has arisen (Strachey, 1959).
1990s	In order to provide well-balanced band-width, telecommunication companies started to build virtual private networks (VPN) through point-to-point electric circuits.
2000s	2002: Amazon web service. 2006: Amazon s3. the first real cloud application. 2007-2008: Google and IBM created cloud service (Sevli, 2011). July 2010: Rackspace Hosting and NASA initiated an open-source cloud software entrepreneur which is known as Open Stack jointly. March 2011: IBM company introduced smart cloud framework. June 2012: Oracle company introduced Oracle Cloud.

BenefitsofCloudcomputing

CloudComputinghasnumerous advantages.Someofthemarelisted below:

- Onecanaccessapplicationsasutilities,overthe Internet.
- Manipulateandconfigurethe applicationonlineatany time.
- It does not require to install a specific piece of software to access or manipulate cloud application.
- CloudComputingoffersonlinedevelopmentanddeploymenttools,programming runtime environment through Platform as a Service model.
- Cloudresourcesareavailableoverthenetworkinamannerthatprovidesplatform independent access to any type of clients.
- CloudComputingofferson-demandself-service.
- Theresources canbeusedwithout interactionwith cloudserviceprovider.

- CloudComputingishighlycosteffectivebecauseitoperatesathigherefficienciewith greater utilization. It just requires an Internet connection.
- CloudComputingoffersloadbalancing thatmakesitmore reliable.



CloudServiceprovider

- Acloudproviderisacompanythatdeliverscloudcomputingbasedservicesandsolutions to businesses and/or individuals.
- This service organization may provide rented and provider-managed virtual hardware, software, infrastructure and other related services.
- Cloud services are becoming increasingly desirable for companies because they offeradvantages in terms of cost, scalability and accessibility.
- Acloudprovider isalsoknown asautilitycomputing provider.
- This role is typically related to that of a managed service provider (MSP), but usually, the latter provides other managed IT solutions.
- Followingarefew famousCloudService Provider

AmazonWebServices	OracleCloud
DigtialOcean	IBMCloud
Rackspace	OpenNebula
LiquidWeb	Pivotal
MicrosoftAzure	CloudSigma
GoogleCloud Platform	Dell Cloud
VMware	LimeStone
Salesforce	Quadranet

Advantagesofcloud computing

- **Lower costs:** Because cloud networks operate at higher efficiencies and with greater utilization, significant cost reductions are often encountered.
- **Easeofutilization:**Dependinguponthetypeofservicebeingoffered,youmayfindthat you do not require hardware or software licenses to implement your service.
- **QualityofService:**TheQualityofService(QoS)issomethingthatyoucanobtainunder contract from your vendor.
- **Reliability:** The scale of cloud computing networks and their ability to provide load balancingandfailovermakesthemhighlyreliable,oftenmuchmorereliablethanwhatyou can achieve in a single organization.
- **Outsourced IT management:** A cloud computing deployment lets someoneelse manage your computing infrastructure while you manage your business. In most instances, you achieve considerable reductions in IT staffing costs.
- **Simplifiedmaintenanceand upgrade:** Becausethesystem is centralized, you can easily apply patches and upgrades. This means your users always have access to the latest software versions.
- **LowBarriertoEntry:**Inparticular,upfrontcapitalexpendituresaredramatically reduced. In cloud computing, anyone can be a giant at any time.

Disadvantagesofcloudcomputing

- **Requiresgoodspeedinternetwithgoodbandwidth:**Toaccessyourcloudservices,you needtohaveagoodinternetconnectionalwayswithgoodbandwidthtouploadordownload files to/from the cloud.
- **Downtime:** Since the cloud requires high internet speed and good bandwidth, there is always a possibility of service outage, which can result in business downtime.
- **Limitedcontrolofinfrastructure:**Sinceyouarenottheoweroftheinfrastructureofthe cloud, hence you don't have any control or have limited access to the cloud infra.
- **Restrictedorlimitedflexibility:**Thecloudprovidesahugelistofservices,butconsuming them comes with a lot of restrictions and limited flexibility for your applications or developments.
- **Ongoing costs:**Although you save your cost of spending on whole infrastructure and its management, on thecloud, you need to keep paying for services as long as you use them. But in traditional methods, you only need to invest once.
- **Security:**Securityofdataisabigconcernforeveryone.Sincethepubliccloudutilizesthe internet, your data may become vulnerable.
- **VendorLock-in:**Althoughthecloudserviceprovidersassureyouthattheywillallowyou toswitchormigratetoanyotherserviceprovider wheneveryouwant,itisaverydifficult process.

- **Technical issues:** Even if you are a tech whiz, the technical issues can occur, and everythingcan'tberesolvedin-house.Toavoidinterruptions,youwillneedtocontactyour service provider for support. However, not every vendor provides 24/7 support to their clients.

Alistofadvantagesanddisadvantagesofcloudcomputing:

Advantagesof Cloud	**DisadvantagesofCloud**
Nocostof infrastructure	Goodinternetconnection&bandwidth required
Minimummanagementandcost	Downtime
Noadministrativeormanagementhassles	Limitedcontrolof infrastructure
Easy accessibility	Restrictedorlimitedflexibility
Payperuse	Ongoing costs
Reliability	Security
Data control	Vendorlock-in
Databackupandrecovery	TechnicalIssues
Hugecloud storage	NA

AssessingtheRoleofOpenStandards

- When you consider the development of cloud computing to date, it is clear that the technology is the result of the convergence of many different standards.
- Cloudcomputingpromiseofscalabilitycompletelychangesthemannerinwhichservices and applications are deployed.
- Without standards, the industry creates proprietary systems with vendor lock-in, sometimesreferredtoas"stovepipe"clouds.Becauseclientsdonotwanttobelockedinto any single system, there is a strong industry push to create standards-based clouds.
- Thecloudcomputingindustryis workingwiththesearchitectural standards:
  - Platformvirtualizationof resources
  - Service-orientedarchitecture
  - Web-applicationframeworks
  - Deploymentofopen-source software
  - StandardizedWeb services
  - Autonomicsystems
  - Grid computing

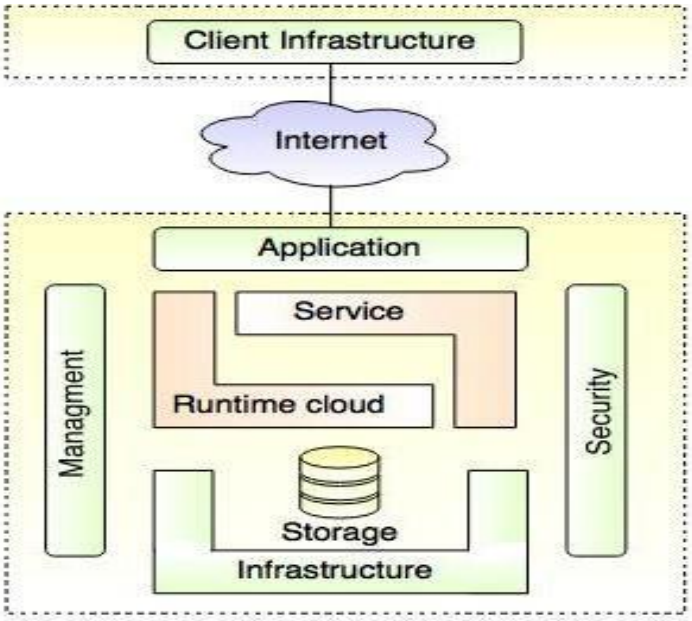
Unit-1
Part-3CloudComputingArchitecture

CloudComputingArchitecture

- Cloudcomputingarchitectureconsistsof manylooselycoupledcloud components.
- **Thearchitectureismainlydividesthecloudarchitectureintotwoparts:**
  1. FrontEnd
  2. BackEnd

  Eachendisconnected toothersthroughanetwork,generallytotheInternet.

Viewofcloudcomputingarchitecture



FrontEnd

- Thefront endis thesideofcomputeruser or client.
- Thecloudtechnologyarchitecturealsoconsistsof**front-end**platformscalledthecloud client which comprises **servers, thin & fat client, tablets & mobile devices**.
- It involves the interfaces and the applications that are necessary to access the CloudComputing system.

BackEnd

- Thebackend isthe cloudsection ofthe system.
- Itinvolvesalltheresourceswhicharenecessary togiveCloudcomputingservices.
- Itincludeshugedatastorage,virtualmachines,securitymechanism,services, deployment models, servers etc.

- To give built-in security mechanism, traffic control and protocols is the responsibility of the back end.
- It is the responsibility of the **back-end** to provide the security of data for cloud users along with the traffic control mechanism.
- Theserveralsoprovidesthemiddlewarewhichhelpstoconnectdevices&communicate with each other.

Cloudcomputingv/straditionalcomputingarchitecture (client/server)

Meaning

- **Cloud** is everywhere and it manages the servers and network infrastructure management. The terms cloud-based refers to anything be it applications, resources or services that is madeavailabletotheusersondemandthroughweb-basedtoolsviatheInternet,asopposed to a direct connection to a server. It's called "cloud computing" because everything from applications to data centers to services is found in the cloud.
- **In Client-Server,** on the other hand, Server is a computer program that provides services to other computer programs and their users. Server based computing refers to the technologybywhichapplicationsgetimplemented,controlled,andoperatedontheserver rather than the client.

Technology

- **Thetermcloud**referstoacollectionofdynamicallyconfiguredsharedresourcesbasedon network technology where each user has access to its own private resource called cloud thatisofferedbyathird-partycloudserviceprovider.Thesecloudserviceprovidersdeliver their computing resources over the internet which can be further accessed through a web browser.
- **InServerbasedcomputing,**ontheotherhand,referstothetechnologywhereadeviceor a program, otherwise known as a server, is designed to managing network resources. The servers accept and respond to requests made by another program, otherwise known as a client.

Application

- A**cloudbased**applicationisanysoftwareprogramorapplicationthatoperatesinthecloud space meaning it's a program running on a cloud infrastructure and can be accessed over the internet by various computing devices through a web browser or a program interface. The cloud applications can be installed on a private cloud.
- A **server based application**, on the other hand, refers to a program or application stored onaremoteserverandaccessedthroughabrowserinterfacesuchasawebbrowser.Servers provide different services such as sharing resources or data among clients along with data access and persistence.

Architecture

- A **cloud computing** architecture is a conceptual model that encompasses all the components and subcomponents required for cloud computing in a cloud space. Cloud provides on-demand access to a networked collection of shared resources like servers, applications, storage, and networks, regardless of where the cloud is.
- **Server architecture**, on the other hand, is the basic foundation on which the server is created or deployed. It basically refers to a network in which clients request and receive service from a centralized server and the server then responds to the requests. It defines how a server along with its components is designed, maintained and managed as a whole.

CloudComputing	Client-Servercomputing
Cloud refers to a shared pool of computing resources that provides on-demand access to these resources via Internet.	Server refers to a dedicated computer which manages access to centralized resources in a network.
Cloud is based on Infrastructure-as-a-Service (IaaS) model that provides virtualized computing resources over the internet.	Server based computing refers to the technology where applications are implementedandcontrolledonthe server.
A cloud based application is any software program or application that operates in the cloud space.	A server based application refers to a program or application stored on a remote server.
Cloud architecture is conceptual model that encompasses all the components and subcomponentsrequired forcloudcomputing	Server architecture is the basic foundationonwhichtheserveriscreated or deployed.

HowCloudComputingWorks?

- Thecloud hasto bedivided intodifferent layers.
- Theselayersarethe**front-end** and**back-end** layers.
- Front-endlayeristhatpartofthecloudwhichuserscaninteract with.
- For example, when we log in to our Gmail account, we see the UI (user interface) where everything works on event-driven buttons and graphics.
- Similarly,asoftwarealsoruns inthefront endofthecloud.
- The back-end includes hardware as well as software that delivers the back-end data from the database to the front end.
- Cloudusesanetworklayertoconnectdifferentdevicestoprovideaccesstoresourcesthat are residing in the centralized data center of the cloud.
- Cloudtechnologyuserscanusethedatacenterthroughthecompany'snetworkorinternet facilities.

- Thistechnologyprovidesvariousadvantages:asuserscanaccessthecloudfrom**anywhere at any time**, but the network bandwidth should have to be more.
- Thistechnologynotonlyfacilitatesdesktopandlaptopusersbutthe**mobileusers**canalso access their business systems based on their demand.
- Aswealreadyknowthatcloudcomputingis**fastandefficient**,applicationsrunningonthe cloud take advantages of flexibility and computing power, i.e., the speed of processing a task.
- Manycomputersofasingleorganizationworktogetheralongwiththeirapplicationonthe cloud as if all the applications were running on a single machine. This flexibility of accessing the cloud resources allows users to use much or little of the resource based on the demand.
- In the Cloud computing system architecture, there is another mechanism of shifting the workload.Localmachinesdon'thavetoperformmassiveliftingoperationswhenitcomes to run applications. Cloud technology can handle those heavy loaded tasks automatically easilyandefficiently.Thisbringsdownthehardware&softwaredemands.Theonlything thattheusershavetothinkisthecloudcomputinginterfacesoftwareofthesystem,which worksmerelyasaweb-browserinthefrontendoftheuser.Thecloud'snetworktakescare of the rest along with the back-end.
- Theback-end is connected throughavirtual networkorinternet. Otherthan that, thereare few more components such as **Middleware, cloud resources**, etc. that includes the cloud computing architecture.
- Thebackendisusedbyserviceprovidersthatinclude**variousservers,computers,virtual machines & data storage facilities** that are combined to form the cloud technology. Its dedicated server handles each application in the system. The front end includes the cloud computing system or network that is used for accessing the cloud computing system. The cloud computing systems' interface varies from cloud to cloud.
- Theback-endhas twoprincipal responsibilities:
    1. Providestrafficcontrol mechanisms,securitypostures& governingtheprotocols
    2. To employ those internet protocols that are connected to the networked computer for communication
- One central server is used to manage the entire cloud system architecture. The server is only responsible for handling the smoothness of traffic without disruption.
- Middleware is a particular type of software that is used to perform processes & also connects networked computers. Depending on the demand of client/user, the storage is provided by the cloud technology's service provider.

ProtocolsusedinCloudcomputing

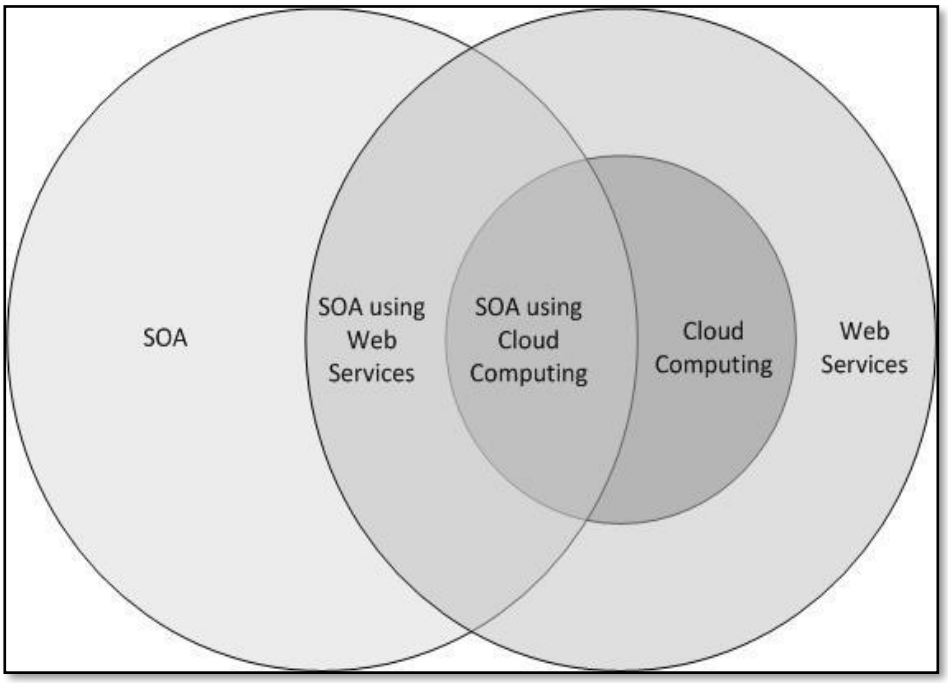
- Cloud computing protocols is a set of rules that allow 2 electronics item to connect and exchange information to one another. Used for storage, communication, for encryption, decryption, networking, security, user-login management etc.

Examplesofcloudcomputing protocols:

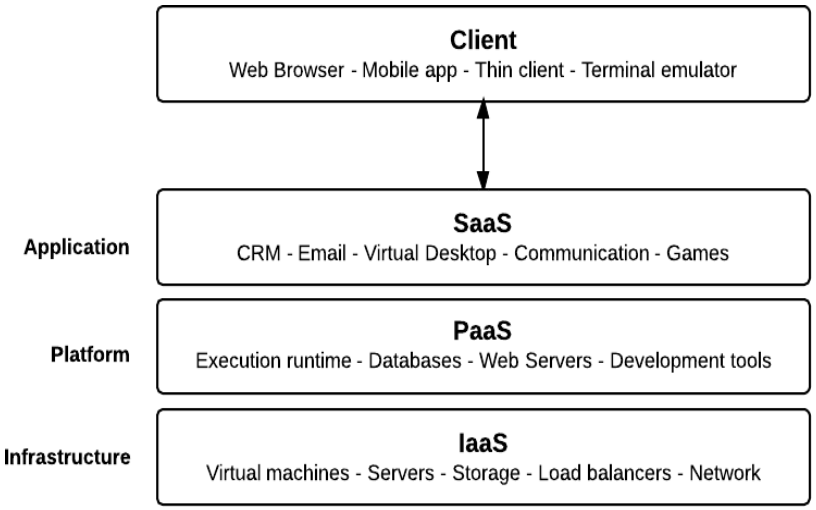
- ✓ Gossip Protocol
- ✓ Connection-lessn/wprotocol(CLNP)
- ✓ StateRoutingProtocol (SRP)
- ✓ InternetGroupManagementProtocol(IGMP)
- ✓ SecureShellprotocol(SSHP)
- ✓ CoverageEnhanced EthernetProtocol(CEE)
- ✓ MediaTransferProtocol(MTP)

RoleofWebServicesinCloudcomputing

- The following diagram illustrates the relationships among Web Services, service-oriented architecture (SOA), and Cloud Computing.
- Web Services encapsulates Cloud Computing in this diagram because Cloud Computing uses Web Services for connections.
- Itispossible,however,touseWebServicesinsituationsotherthanCloudComputing.
- WebServices couldbe simplybeaconnection.
- It is possible to have a service-oriented architecture and not use Web Services for connections.
- ThetechnologyofCloudcomputingentailstheconvergenceofGridandclustercomputing, virtualization, Web services and Service Oriented Architecture (SOA) - it offers the potential to set ITfree from thecosts and complexity ofits typical physical infrastructure, allowing concepts such as Utility Computing to become meaningful.
- Key players: IBM, HP, Google, Microsoft, Amazon Web Services, Salesforce.com, NetSuite, VMware.



ServiceModelsofCloudComputing



1. Software-as-a-Service(SaaS)

- SaaSisknownas'On-DemandSoftware'.
- Itisasoftwaredistributionmodel.Inthismodel,theapplicationsarehostedbyacloud service provider and publicized to the customers over internet.
- InSaaS,associateddata andsoftwarearehostedcentrallyonthecloudserver.
- UsercanaccessSaaSbyusingathinclientthroughaweb browser.
- SAAS has become an increasingly widespread delivery model as underlying technologiesthatsupportWebservicesandservice-orientedarchitecture(SOA)mature and new development approaches, such as Ajax, become popular.
- CRM,OfficeSuite,Email,games,etc.arethesoftwareapplicationswhichareprovided as a service through Internet.
- ThecompanieslikeGoogle,Microsoftprovidetheirapplicationsasaservicetotheend users.

AdvantagesofSaaS

- SaaSis**easytobuy**becausethepricingofSaaSisbasedonmonthlyorannualfeeand it allows the organizations to access business functionalities at a small cost, which is less than licensed applications.
- SaaS **needed less hardware**, because the software is hosted remotely, hence organizations do not need to invest in additional hardware.
- **Less maintenance cost** is required for SaaS and do not require special software or hardware versions.
- **Easier administration.**
- **Automaticupdatesandpatchmanagement.**
- **Compatibility:**alluserswillhavethesameversionofsoftware.
- **Globalaccessibility.**

DisadvantagesofSaaS

- SaaS applications are totally dependent on Internetconnection. They are not usablewithout Internet connection.
- ItisdifficulttoswitchamongsttheSaaSvendors.

2. Platform-as-a-Service(PaaS)

- PaaSisaprogrammingplatformfordevelopers.Thisplatformisgeneratedforthe programmers to create, test, run and manage the applications.
- AdevelopercaneasilywritetheapplicationanddeployitdirectlyintoPaaSlayer.
- PaaSgivestheruntimeenvironmentforapplicationdevelopmentanddeploymenttools.
- PAASfreesusersfromhavingtoinstallin-househardwareandsoftwareto developor run a new application.
- PAASdoesn'treplacea business'entireinfrastructurebutinstead,abusinessrelieson PAAS providers for key services, such as Java development or application hosting.
- PAASproviders thenchargeforthataccesson aper-user basisor onamonthly basis.
- Users only need to log in and start using the platform-usually through a Web browser interface.
- **Google Apps Engine(GAE),** Windows Azure, SalesForce.com are the examples ofPaaS.

AdvantagesofPaaS

- PaaSiseasiertodevelop.Developercanconcentrateonthedevelopmentand innovation without worrying about the infrastructure.
- InPaaS,developeronlyrequiresaPCandanInternetconnectiontostartbuilding applications.
- ProvidesSecurityandredundancy.
- Buildanddeploymenttoolsforrapidapplicationmanagementanddeployment.
- Integrationwithotherinfrastructurecomponentssuchaswebservicesanddatabases
- Multi-tenancy,platform servicethat canbeused bymanyconcurrentusers.
- Logging,reporting, andcodeinstrumentation.

DisadvantagesofPaaS

- Onedeveloper can write theapplicationsaspertheplatformprovidedbyPaaSvendor hence the moving the application to another PaaS vendor is a problem.

3. Infrastructure-as-a-Service(IaaS)

- IaaS is a way to deliver acloud computing infrastructure like server, storage, network and operating system.
- The customers can access these resources over cloud computing platform i.e. Internet as an on-demand service.
- InIaaS,youbuycompleteresourcesratherthanpurchasingserver,software,datacenter space or network equipment.

- IaaSwasearliercalledasHardwareasaService(HaaS).ItisaCloudcomputing platform based model.
- HaaSdiffersfromIaaSinthewaythatusershavethebarehardwareonwhichtheycan deploy their own infrastructure using most appropriate software.

AdvantagesofIaaS

- In IaaS,usercandynamicallychooseaCPU,memorystorageconfigurationaccording to need.
- Userscaneasilyaccessthevastcomputingpower availableon IaaSCloud platform.

DisadvantagesofIaaS

- IaaScloudcomputingplatformmodelisdependentonavailabilityofInternetand virtualization services.

DeploymentModels

There are four main cloud deployment models that differ significantly and for which most of the companies opt: a public, private, hybrid and a community one

1. **Privatecloud:**
   - Thecloudinfrastructureisprovisionedforexclusiveusebya**singleorganization** comprisingmultiple consumers.
   - It may be owned, managed, and operated by the organization, a third party, or somecombination of them, and it may exist on or off premises.
   - The**advantages** ofusingaprivatecloudare:
     1. Individualdevelopment
     2. Storageandnetwork componentsare customizable
     3. Highcontroloverthecorporate information
     4. Highsecurity,privacyand reliability
   - ThePrivatecloud hasthe following**disadvantages**:
     1. **Poorscalability**:Privatetypeofcloudsisscaledwithininternallimitedhosted resources.
     2. **Costly:**Asitprovidessecuredandmorefeatures,soit'smoreexpensivethana public cloud.
     3. **Pricing:**isinflexible;i.e.,purchasingnewhardwareforup-gradationismore costly.
     4. **Restriction:** It can be accessed locally within an organization and is difficult toexpose globally.

2. **PublicCloud**
   - Public cloud is a type of cloud hosting that allows the accessibility of systems & itsservices to its clients/users easily.
   - Someof theexamplesof thosecompanieswhichprovide public cloudfacilitiesare IBM, Google, Amazon, Microsoft, etc.

- Thiscloud serviceisopen foruse.
- Publiccloudsareavailable tothe generalpublicanddata arecreated andstoredon **third-party servers**.
- As server infrastructure belongs to service providers that manage them and administer pool resources, the need for user companies to buy and maintain their own hardware is eliminated.
- Provider companies offer resources as a service on a free of charge or pay-per-use basis via the Internet connection.
- At the same time, relying on a third party in running their infrastructure deprives users of knowing where their information is kept and who has access to it.
- The**advantages**ofthePubliccloud are:
  1. Flexible
  2. Reliable
  3. High Scalable
  4. Lowcost
  5. Placeindependence
- Thistypealsoholds some **disadvantages**such as:
  1. LessSecured
  2. Poor Customizable

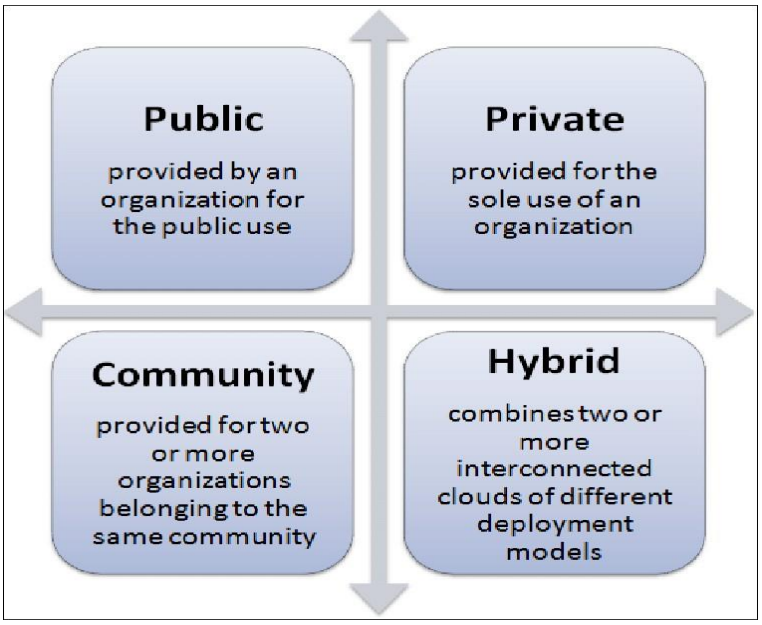
3. CommunityCloud

- Community cloud is another type of cloud computing in which the setup of the cloud is shared manually among different organizations that belong to the same community or area.
- Exampleofsuchacommunityiswhereorganizations/firmsaretherealongwiththe financial institutions/banks.
- For joint business organizations, ventures, research organizations and tenders community cloud is the appropriate solution.
- The**advantages**of acommunitycomputing typeincludethe following:
  1. Cost reduction
  2. Improvedsecurity,privacyandreliability
  3. Easeofdatasharingandcollaboration
- Thedisadvantages are:
  1. Highercostthan thatofa publicone
  2. Sharingoffixedstorage andbandwidth capacity
  3. Itisnotwidespreadsofar

4. Hybridcloud:

- **Hybrid Cloud** is integrated, i.e., it can be a combination of two or more cloud servers,i.e.,private,publicorcommunitycombinedasonearchitecture,butremain individual entities.

- Non-criticaltaskssuchasdevelopmentandtestworkloadscanbedoneusingpublic cloudwhereascriticaltasksthataresensitivesuchasorganizationdatahandlingare done using a private cloud.
- In this Benefits of both public and private deployment models, as well as a community deployment model, are possible in a hybrid cloud hosting.
- Itcancrossisolationandovercomeboundariesbytheprovider;hence,itcannotbe simplycategorizedintoanyofthethreedeployments-public,privateorcommunity cloud.
- Advantagesof HybridCloudComputingare:
  1. Flexible
  2. Secure
  3. Cost Effective
  4. Rich Scalable
- DisadvantagesofHybrid Cloudare:
  1. Complexnetworkingproblem
  2. Organization'ssecurity Compliance



Unit–2
Part-1InfrastructureasaService(IaaS)

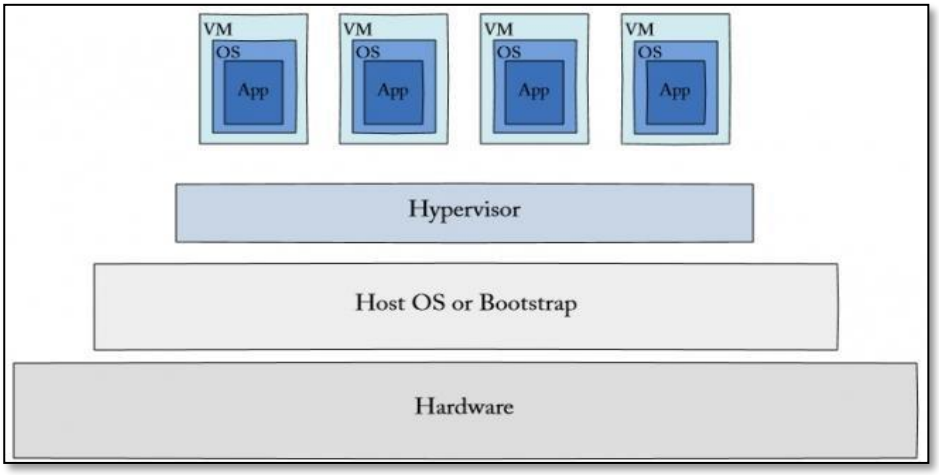
IntroductiontoIaaS

- Cloud infrastructure services, known as "Infrastructure as a Service" (IaaS), **deliver computer infrastructure, storage, and networking**.
- **IaaS**, as the name suggests, is a way of providing Cloud computing infrastructure such as virtual machines, storage drives, servers, operating systems & networks, which is also an on-demand service.
- Insteadofpurchasingserversordevelopingsoftware,clientsbuythoseresourcesasafully outsourced service based on their requirement.
- "Publiccloud"isconsideredasaninfrastructurethatconsistsofsharedresources,basedon a self-service over the Internet.
- Inoneword,itistheonlylayerofthecloudwherethecustomergetstheplatformfortheir organization to outsource IT infrastructure on a pay-per-use basis.
- IaaSprovidesuserswith:
  1. Load balancers
  2. Diskstorageviavirtualmachines
  3. SoftwarePackages
  4. IPaddress
  5. VLANs
- AdvantagesofIaaS are:
  1. **Dynamic:**Userscandynamicallyopt&configuredevicessuchasCPU,storagedrive, etc.
  2. **EasyAccess:**Userscaneasilyaccessthevastcloudcomputing power.
  3. **Renting:**Flexibleand efficientwhilerenting IT infrastructures.
  4. **Fullcontrol** ofcomputerresourcesalongwithportability.
- Disadvantagesof IaaSareasfollows:
  1. Internetconnectionisa must.
  2. IaaSdependsonvirtualizationservices.
  3. Thisservicerestrictsuser-privacy&customization

Introductiontovirtualization

- Virtualizationistheabilitywhichallowssharingthephysicalinstanceofasingle application or resource among multiple organizations or users.
- Thistechniqueisdonebyassigninganamelogicallytoallthosephysicalresources& provides a pointer to those physical resources based on demand.

- Overanexistingoperatingsystem&hardware,wegenerallycreateavirtualmachinewhich and above it we run other operating systems or applications. This is called **Hardware Virtualization.**
- The virtual machine provides a separate environment that is logically distinct from its underlying hardware.
- Here, the system or the machine is the host & virtual machine is the guest machine. This virtual environment is managed by a firmware which is termed as a **hypervisor.**



- Therare**severalapproaches**orways tovirtualizescloudservers. Theseare:
  - **GridApproach**:wheretheprocessingworkloadsaredistributedamongdifferent physical servers, and their results are then collected as one.
  - **OS -Level Virtualization**: Here, multiple instances of an application can run in anisolated form on a single OS
  - **Hypervisor-basedVirtualization**:whichiscurrently themostwidelyused technique
- With hypervisor's virtualization, there arevarious sub-approaches to fulfill the goal to run multiple applications & other loads on a single physical host.
- A technique is used to allow virtual machines to move from one host to another without any requirement of shutting down. This technique is termed as **"Live Migration"**.
- Another technique is used to actively load balance among multiple hosts to efficiently utilize those resources available in a virtual machine, and the concept is termed as **Distributed Resource Scheduling or Dynamic Resource Scheduling.**

TypesofVirtualization

- The virtualization of cloud has been categorized into four different types based on theircharacteristics. These are:
  1. HardwareVirtualization
  2. Full Virtualization
  3. EmulationVirtualization
  4. Para-virtualization
  5. SoftwareVirtualization

6. OS Virtualization
7. ServerVirtualization
8. StorageVirtualization

HowVirtualizationworksincloud?

- Virtualization plays a significant role in cloud technology and its working mechanism. Usually,whathappensinthecloud-theusersnotonlysharethedatathatarelocatedinthe cloudlikeanapplicationbutalsosharetheirinfrastructureswiththehelpofvirtualization.
- Virtualization is used mainly to provide applications with standard versions for the cloud customers & with the release of the latest version of an application the providers can efficiently provide that application to the cloud and its users and it is possible using virtualization only.
- By the use of this virtualization concept, all servers & software other cloud providers require those are maintained by a third-party, and the cloud provider pays them on a monthly or yearly basis.
- In reality, most of the today's hypervisor make use of a combination of different types of hardwarevirtualization.Mainlyvirtualizationmeansrunningmultiplesystemsonasingle machinebutsharingallresources(hardware)&ithelpstoshareITresourcestogetbenefit in the business field.

Advantagesofvirtualization:

- Thenumberofservers gets reducedby the useof virtualization concept
- Improvetheabilityoftechnology
- Thebusiness continuity alsoraised duetotheuseof virtualization
- Itcreatesamixedvirtualenvironment
- Increaseefficiencyfordevelopment&test environment
- LowersTotalCost ofOwnership (TCO)

Featuresofvirtualization:

1. Partitioning:Multiplevirtualservers canrunona physicalserveratthesame time
2. Encapsulationofdata:Alldataonthevirtualserverincludingbootdisksisencapsulated in a file format
3. Isolation:TheVirtualserverrunningonthephysicalserveraresafelyseparated&don't affect each other
4. Hardware Independence: When the virtual server runs, it can migrate to the different hardware platform

Differentapproachestovirtualization

- Server virtualization can be viewed as a part of overall virtualization trend in the IT companies that include network virtualization, storage virtualization & management of workload.
- Thistrend bringsdevelopment inautomatic computing.

ForServerVirtualization,therearethreepopularapproaches.These are:

1. VirtualMachinemodel
2. Para-virtualMachinemodel
3. OperatingSystem(OS)layerVirtualization

1. **VirtualMachine model:**
   - Itisbasedonhost-guestparadigm,whereeachguestrunsonavirtualreplicaof hardware layer.
   - Thistechnique ofvirtualizationprovide guest OSto runwithout modification.
   - Howeveritrequiresrealcomputingresourcesfromthehostandforthisahypervisoror VM is required to coordinate instructions to CPU.
2. **Para-VirtualMachine model:**
   - Itisalso basedon host-guestparadigm& usesvirtual machinemonitor too.
   - InthismodeltheVMMmodifiestheguestoperatingsystem'scodewhichiscalled 'porting'.
   - LikethatofvirtualmachineSimilarlythePara-virtualmachineisalsocapableof executing multiple operating systems.
3. **OperatingSystem Layer Virtualization**:
   - Virtualization at OS level functions in a different way and is not based on host-guest paradigm.
   - Inthismodelthehostrunsasingleoperatingsystemkernelasitsmain/coreand transfers its functionality to each of the guests.
   - Theguestmust usethe sameoperatingsystem asthehost.
   - Thisdistributednatureofarchitectureeliminatedsystemcallsbetweenlayersandhence reduces overhead of CPU usage.
   - Itisalsoamustthateachpartitionremainsstrictlyisolatedfromitsneighborsbecause anyfailureorsecuritybreachofonepartitionwon'tbeabletoaffecttheotherpartitions.

ServerVirtualization

- It is the division of physical server into several virtual servers and this division is mainly done to improvise the utility of server resource.
- In other word it is the masking of resources that are located in server which includes the number & identity of processors, physical servers & the operating system.
- Thisdivisionofonephysicalserverintomultipleisolatedvirtualserversisdonebyserver administrator using software.
- Thevirtualenvironmentissometimescalledthe**virtualprivate-servers**.
- Inthisprocess,theserverresourcesarekepthiddenfromtheuser.
- Thispartitioningofphysicalserverintoseveralvirtualenvironments;resultinthe dedication of one server to perform a single application or task.

UsageofServerVirtualization

- Thistechniqueismainlyusedinweb-serverswhichreducesthecostofweb-hosting services.
- Instead ofhaving separatesystem for each web-server,multiplevirtual serverscan runon the same system/computer.

Theprimaryusesofservervirtualizationare:

- Tocentralizetheserver administration
- Improvetheavailabilityofserver
- Helpsindisasterrecovery
- Easeindevelopment&testing
- Makeefficientuseofserver resources.

Advantages:

- **CostReduction:**Server virtualizationreducescostbecauselesshardwareis required.
- **Independent Restart:** Each server can be rebooted independently and that reboot won't affect the working of other virtual servers.

TypesofVirtualization:Resourcevirtualization

1. HardwareVirtualization.
2. Operatingsystem Virtualization.
3. ServerVirtualization.
4. StorageVirtualization.

1. HardwareVirtualization:

- Whenthevirtualmachinesoftwareorvirtualmachinemanager(VMM)isdirectly installed on the hardware system is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory andother hardware resources.
- After virtualization of hardware system we can install different operating system on it and run different applications on those OS.

Usage:

- Hardware virtualization is mainly done for the server platforms, because controllingvirtual machines is much easier than controlling a physical server.

2. OperatingSystemVirtualization:

- Whenthevirtualmachinesoftwareorvirtualmachinemanager(VMM)isinstalledon the Host operating system instead of directly on the hardware system is known as operating system virtualization.

Usage:

- OperatingSystemVirtualizationismainlyusedfortestingtheapplicationsondifferent platforms of OS.

3. ServerVirtualization:

- Whenthevirtualmachinesoftwareorvirtualmachinemanager(VMM)isdirectly installed on the Server system is known as server virtualization.

Usage:

- Servervirtualizationisdonebecauseasinglephysicalservercanbedividedinto multiple servers on the demand basis and for balancing the load.

4. StorageVirtualization:

- Storagevirtualizationistheprocessofgroupingthephysicalstoragefrommultiple network storage devices so that it looks like a single storage device.
- Storagevirtualizationisalsoimplementedbyusingsoftwareapplications.

Usage:

- Storagevirtualizationismainlydoneforback-upandrecoverypurposes.

WhatdoesVirtualProvisioningmean?

- Virtual provisioning is a virtual storage network (VSAN)-based technology in which storage space is allocated on demand to devices.
- Thisprocessallowsvirtualizedenvironmentstocontroltheallocationand managementof physical disk storage connected with virtual machines (VM).
- Virtualprovisioningisalsoknownas **thinprovisioning**.However,virtualprovisioningis morerelevanttoavirtualenvironment,whilethinprovisioningismorerelevanttophysical computing implementations.

Hypervisor

- A**hypervisor**, also known as avirtual machine monitor,is aprocess that creates and runs virtual machines (VMs).
- AhypervisorallowsonehostcomputertosupportmultipleguestVMsbyvirtuallysharing its resources, like memory and processing.
- Therearetwotypes of hypervisors.
- Type1hypervisors,called**"baremetal,"**rundirectlyonthehost'shardware.
- Type 2 hypervisors, called **"hosted,"** run as a software layer on an operating system, like other computer programs.
- Hypervisors make it possible to use more of a system's available resources and provide greaterITmobilitysincetheguestVMsareindependentofthehosthardware.Thismeans they can be easily moved between different servers.

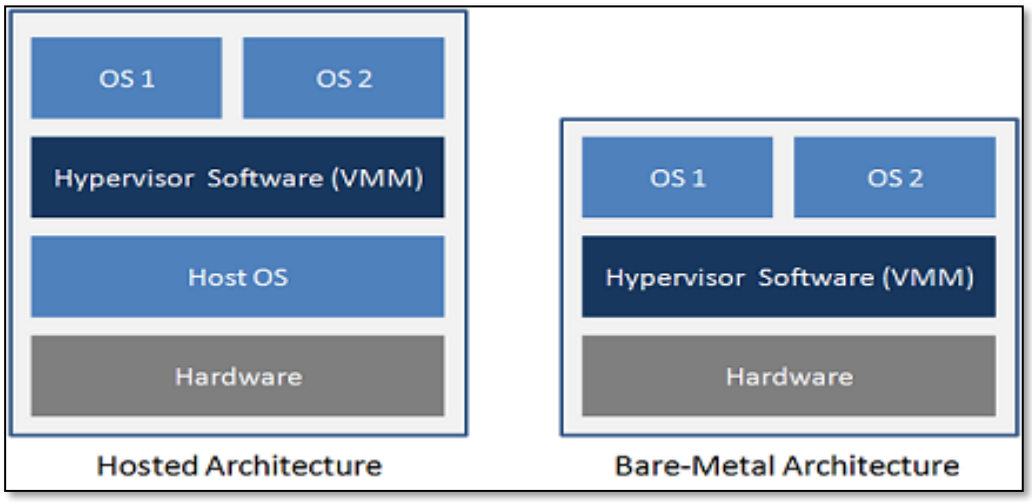
TYPE-1 Hypervisor:

- Hypervisorrunsdirectlyonunderlyinghost system.
- Itisalsoknown as"**NativeHypervisor"or"Baremetalhypervisor**".
- Itdoesnotrequireanybaseserveroperatingsystem.

- Ithasdirectaccesstohardwareresources.
- ExamplesofType1hypervisorsinclude**VMwareESXi,CitrixXenServerand Microsoft Hyper-V hypervisor.**

TYPE-2 Hypervisor:

- AHostoperating systemrunson underlyinghost system.
- Itisalsoknown as **'HostedHypervisor".**
- Basicallyasoftwareinstalledonanoperatingsystem.
- Hypervisorasksoperatingsystem tomakehardware calls.
- ExampleofType2hypervisorincludeVMwarePlayer orParallels Desktop.
- Hostedhypervisors areoftenfoundonendpoints likePCs.



MachineImage

- Machine imaging is a process that is used to achieve the goal of system portability, provision, and deploy systems in the cloud through capturing the state of systems using a system image.
- Asystem imagemakesacopyoracloneofthe entirecomputersystem insideasinglefile.
- Theimage is made by using aprogram called system imaging program and can beused later to restore a system image.
- For example Amazon Machine Image (AMI) is a system image that is used in the cloudcomputing.
- TheAmazonWebServices usesAMIto storecopies ofavirtualmachine.
- AnAMIisafilesystemimagethatcontainsanoperatingsystem,alldevicedrivers,and any applications and state information that the working virtual machine wouldhave.
- TheAMIfilesareencryptedandcompressedforsecuritypurposeandstoredinAmazon S3 (Simple Storage System) buckets as a set of 10MB chunks.
- Machine imaging is mostly run on virtualization platform due to this it is also called as Virtual Appliances and running virtual machines are called instances.

- Because many users share clouds, the cloud helps you track information about images, such as ownership, history, and so on.
- TheIBMSmartCloudEnterpriseknowswhatorganizationyoubelongtowhenyoulogin.
- Youcanchoosewhethertokeepimagesprivate,exclusivelyforyourownuse,orto share with other users in your organization.
- Ifyouareanindependentsoftwarevendor,youcanalsoaddyourimagestothepublic catalog.

VirtualMachine

- **Avirtualmachine**,knownasaguest,iscreatedwithinacomputingenvironment,calleda host.
- Multiplevirtualmachinescan existinone hostat onetime.
- Virtual machines are software computers that provide the same functionality as physical computers.
- Likephysicalcomputers, theyrunapplicationsandanoperatingsystem.
- However,virtual machinesare computer filesthat runon a physical computerand behave like a physical computer.
- Inotherwords,virtualmachinesbehaveasseparatecomputersystems.
- Virtualmachinescan alsobeused forotherpurposes suchasserver virtualization.
- Specializedsoftware,calledahypervisor, emulatesthePC clientorserver'sCPU, memory,hard disk, network and other hardware resources completely, enabling virtual machines to share the resources.

AdvantagesofVirtualMachines:

- Providesdisasterrecovery andapplicationprovisioning options
- Virtualmachinesaresimplymanaged,maintained, andarewidely available
- Multipleoperatingsystemenvironments canbe runon asinglephysical computer

DisadvantagesofVirtualMachines:

- Running multiple virtual machines on one physical machine can cause unstable performance
- Virtualmachinesarelessefficientandrunslower thanaphysicalcomputer.

TypesofVirtualMachines:

1. **Process virtual machines**: Execute computer programs in a platform-independent environment. It masks the information of the underlying hardware or operating system. This allows the program to be executed in the same fashion on any platform.
2. **System virtual machines**: Support the sharing of a host computer's physical resources between multiple virtual machines.

Datastorageincloudcomputing(storageasaservice)

Introduction

- Cloudstorageis a servicewhich enablessaving thedata on offsidestoragesystem.
- Thisdataismanagedbythird-party.
- Thisdataisaccessible byawebservices API.
- **CloudStorage**istechnologythatallowsyoutosavefilesinstorage,andthenaccessthose files via the Cloud.
- Let's break down this definition. First, storage is the computer's ability to save files andother resources for later use.
- Whenyourestartacomputer,thefilesthatarestillavailableafterthecomputerturnsback on are saved and read from storage.
- Suchstorage commonly consistsofa harddrive,a USBFlashdrive,oranother typeof drive.
- Becauselocaldatadrivescanbedamagedorstolen,anideawasdevelopedtousedata drives over a network as storage.
- Thisallowsthedrivesto besecured inadatacenterand backedupautomatically.
- Initially,networkstoragerequiredfastlocalnetworks(LAN),buttodaywehaveaubiquitous network called the Internet.
- ThesecondpartofCloudStorage,theCloud,representstheInternet.Anyservice,including storage, available over the Internet, is called Cloud service.
- For ex: GMAIL it is email in the Cloud and An Amazon MP3 player, that's music in the Cloud.

StorageDevices

Followingarethe categoriesofstorage devices:

1) **BlockStorageDevices**–Thistypeofdevicesproviderawstoragetotheclients.Thisraw storageisseparatedforcreatingvolumes.Avolumeisarecognizableunit ofdatastorage.
2) **File Storage Devices –** The file storage devices are provided to the client in the form of files for maintaining its file system. Storage data is accessed using the Network File System(NFS).

StorageClassesofcloud

Followingarethe categoriesofstorage classes:

1. **UnmanagedCloudStorage**
   - Thestorageispreconfiguredforthecustomer,thisisknownas**unmanagedcloud storage.**
   - Thecustomercannot formatorinstallhis ownfilesystemor changedrive properties.
2. **ManagedCloudStorage**
   - Managedcloud storageprovides the onlinestoragespaceon-demand.
   - Thissystem shows theuser likeraw diskthat theusercan partition and format.

AmazonElasticComputeCloud(AmazonEC2)

- **AmazonEC2(ElasticComputeCloud)**isawebserviceinterfacethatprovidesresizable compute capacity in the AWS cloud.
- It is designed for developers to have complete control over web-scaling and computingresources.
- UsingAmazonEC2eliminatesyourneedtoinvestinhardwareupfront,soyoucandevelop and deploy applications faster.
- YoucanuseAmazonEC2tolaunchasmanyorasfewvirtualserversasyouneed, configure security and networking, and manage storage.
- AmazonEC2enablesyoutoscaleupordowntohandlechangesinrequirementsorspikes in popularity, reducing your need to forecast traffic.

FeaturesofAmazonEC2

Amazon EC2 provides thefollowing features:

- Virtualcomputingenvironments,knownas**instances**
- Preconfigured templates for your instances, known as Amazon Machine Images (AMIs), that package the bits you need for your server
- VariousconfigurationsofCPU,memory,storage,andnetworkingcapacityforyour instances, known as **instance types**.
- Secureloginformationforyourinstances usingkey pairs.
- Storagevolumesfortemporarydatathat'sdeletedwhenyoustoporterminateyourinstance, known as **instance store volumes**
- PersistentstoragevolumesforyourdatausingAmazonElasticBlockStore,knownas **AmazonEBS volumes**
- Multiplephysicallocationsforyourresources,suchasinstancesandAmazonEBSvolumes, known as **Regions and Availability Zones**
- A firewall that enables you to specify the protocols, ports, and source IP ranges that canreach your instances using security groups
- StaticIPv4addressesfor dynamiccloudcomputing,knownasElastic IPaddresses
- Metadata,knownas tags,thatyoucancreateandassigntoyourAmazonEC2 resources
- VirtualnetworksyoucancreatethatarelogicallyisolatedfromtherestoftheAWScloud, andthatyoucanoptionallyconnecttoyourownnetwork,knownas *virtualprivateclouds* (VPCs)

PricingforAmazonEC2

- WhenyousignupforAWS,youcangetstartedwithAmazonEC2forfreeusingtheAWSFree Tier
- Generally,AmazonEC2pricedonperinstance/perhour basis.
- However, any instance can be rented on per month basis as well. In such case, Reserved and Spot Instances pricing can be applied resulting in significant discount. Instances are priced depending on their "size", namely how much CPU and RAM included.

- AmazonEC2 provides thefollowing purchasing options for instances:
  1. **On-Demand Instances**
     Pay for the instances that you use by the second, with no long-term commitments orupfront payments.
  2. **SavingsPlans**
     YoucanreduceyourAmazonEC2costsbymakingacommitmenttoaconsistent amount of usage, in USD per hour, for a term of 1 or 3 years.
  3. **ReservedInstances**
     YoucanreduceyourAmazonEC2costsbymakingacommitmenttoaspecificinstance configuration, including instance type and Region, for a term of 1 or 3 years.
  4. **Spot Instances**
     RequestunusedEC2instances,whichcanreduceyourAmazonEC2costssignificantly.

AmazonEC2ComputeUnit

- Amazon'sEC2inadditiontobeingoneoftheoldestandmostmatureclouds server platforms, also provides clearly defined CPU tiers across its 8 different instance sizes.
- These are defined in terms of ECUs (EC2 Compute Unit) where 1 ECU is the equivalent CPU capacity of a 1.0-1.2 GHz 2007 Opteron or 2007 Xeon processor.
- Theirinstancesizes includesthe following:
  - ✓ Small/m1.small(32-bit)=1 ECU
  - ✓ Large/m1.large=4ECUs
  - ✓ High-CPUMedium/c1.medium(32-bit)=5 ECUs
  - ✓ High-MemoryExtraLarge/m2.xlarge=6.5 ECUs
  - ✓ ExtraLarge/m1.xlarge = 8ECUs
  - ✓ High-MemoryDoubleExtraLarge/m2.2xlarge = 13 ECUs
  - ✓ High-CPUExtraLarge/c1.xlarge=20ECUs
  - ✓ High-MemoryQuadrupleExtraLarge/m2.4xlarge=26 ECUs

StorageofAmazon EC2

Economical,High-ScaleStorage Choices

1. **Amazon Glacier-** It is an extremely low-cost storage service. It offers secureand fast storage for data archiving and backup.
2. **Amazon Elastic Block Store (EBS)-** It provides block-level storage to use with AmazonEC2instances.AmazonElasticBlockStorevolumesarenetwork-attachedand remain independent from the life of an instance.
3. **AWS Storage Gateway-** This AWS service is connecting on-premises software applications with cloud-based storage. It offers secure integration between the company's on-premises and AWS's storage infrastructure.

Companiesusing AWS

- Instagram
- Zoopla
- Pinterest
- Netflix
- Dropbox
- Etsy
- Talkbox
- Playfish
- Ftopia

Eucalyptus

- Eucalyptus is open source software for building AWS-compatible private and hybrid clouds.
- AsanInfrastructureasaService(IaaS)product,Eucalyptusallowsyouruserstoprovision your compute and storage resources on-demand.
- Amazon has partnered with Eucalyptus to deliver an Infrastructure as a Service (IaaS) product.
- As part of the deal, Amazon Web Services (AWS) is letting the IaaS provider tap into its application programming interfaces (APIs).
- Eucalyptus says it is now "fully compatible with the Amazon Web Services API, which means you can use or reuse your existing AWS-compatible tools, images (AMIs), and scripts to manage your own hybrid and on premise clouds."
- Eucalyptus'sIaaSserviceisnowcompatiblewiththefollowingAWSservices:EC2,EBS, AMI, S3, and IAM.

Unit2
Part-2CloudSecurity

InfrastructureSecurity

- Cloud computing utilizes three delivery models (SaaS, PaaS, and IaaS) to provide infrastructure resources, application platform and software as services to the consumer.
- Theseservicemodels needdifferentlevel ofsecurityin thecloud environment.
- Cloud service providers and customers are responsible for security and privacy in cloud computing environments but their level of responsibility will differ for different delivery models.
- Infrastructure as a Service (IaaS) serves as the foundation layer for the other delivery models, and a lack of security in this layer affects the other delivery models.
- InIaaS,althoughcustomersareresponsibleforprotectingoperatingsystems,applications, andcontent,thesecurityofcustomerdataisasignificantresponsibilityforcloudproviders.
- In Platform as a service (PaaS), users are responsible for protecting the applications that developers build and run on the platforms, while providers are responsible for taking care of the users' applications and workspaces from one another.
- InSaaS,cloudproviders,particularlypubliccloudproviders,havemoreresponsibilitythan clientsforenhancingthesecurityofapplicationsandachievingasuccessfuldatamigration.
- IntheSaaSmodel,databreaches,applicationvulnerabilitiesandavailabilityareimportant issues that can lead to financial and legal liabilities.

INFRASTRUCTURESECURITY:THENETWORKLEVEL

- As network level of infrastructure security is concerned, it is important to distinguish between public clouds and private clouds.
- Withprivateclouds,therearenonewattacks,vulnerabilities,orchangesinriskspecificto this topology that information security personnel need to consider.
- Ifpubliccloudservices,changingsecurityrequirementswillrequirechangestothenetwork topology and the manner in which the existing network topology interacts with the cloud provider's network topology should be taken into account.
- Alldataon the networkneed to be secured.
- Strong network traffic encryption techniques such as Secure Socket Layer (SSL) and the Transport Layer Security (TLS) can be used to prevent leakage of sensitive information.
- Several key security elements such as data security, data integrity, authentication and authorization, data confidentiality, web application security, virtualization vulnerability,

availability,backup,anddatabreachesshouldbecarefullyconsideredtokeepthecloudup and running continuously.

- Therearefoursignificantriskfactorsinthisusecase:
  1. Ensuring the confidentiality and integrity of organization's data-in-transit to and from a public cloud provider
  2. Ensuringproperaccess control
  3. Ensuringtheavailabilityofthe Internet-facingresources
  4. Replacingtheestablishedmodelofnetworkzonesandtierswithdomains.

INFRASTRUCTURESECURITY-THEHOSTLEVEL

- When reviewing host security and assessing risks, the context of cloud services delivery models(SaaS,PaaS,andIaaS)anddeploymentmodelspublic,private,andhybrid)should be considered.
- ThehostsecurityresponsibilitiesinSaaSandPaaSservicesaretransferredtotheprovider of cloud services.
- IaaScustomersareprimarilyresponsibleforsecuringthehostsprovisionedinthecloud in terms virtualization software security, customer guest OS or virtual server security.

INFRASTRUCTURESECURITY-THEAPPLICATIONLEVEL

- Softwaresecurityor applicationsshould beacrucial elementof asecurity program.
- Most enterprises with information security programs have yet to introduce an application security program to address this domain.
- Mostwebsitesaresecuredatthenetworklevelwhiletheremaybesecurityloopholesatthe application level which may allow information access to unauthorized users.
- Software and hardware resources can be used to provide security to applications. In this way, attackers will not be able to get control over these applications and change them.
- XSS attacks, Cookie Poisoning, Hidden field manipulation, SQL injection attacks, and GoogleHackingaresomeexamplesofthreatstoapplicationlevelsecuritywhichresulting from the unauthorized usage of the applications.
- Designing and implementing applications aims at deployment on a cloud platform will requireexistingapplicationsecurityprogramstoreexaminecurrentpracticesandstandards. The application security spectrum ranges from single-user applications to multiuser e-commerce applications used by many users.

Thelevelisresponsiblefor managing

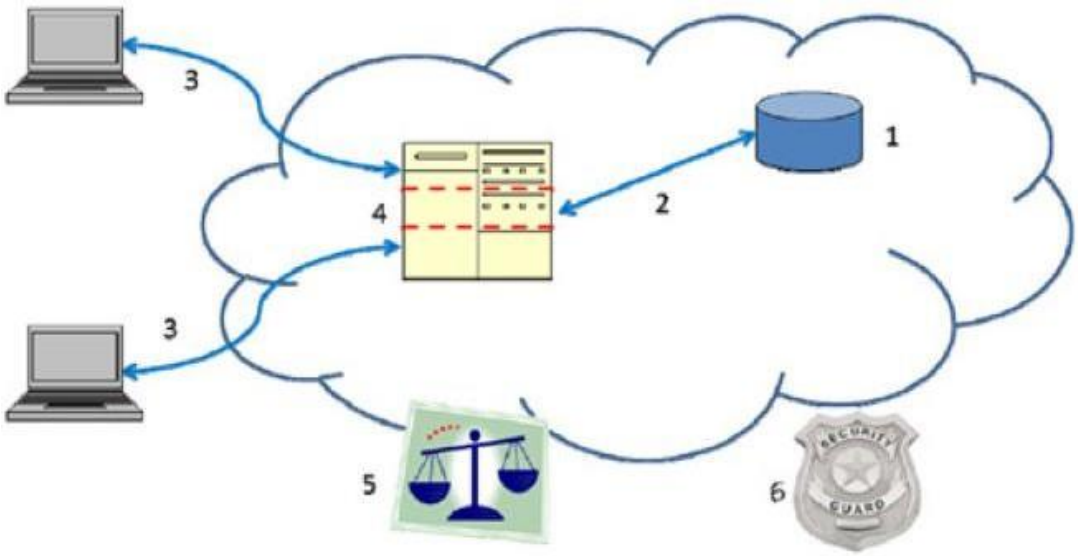
- Application-levelsecurity threats;
- Enduser security;
- SaaSapplicationsecurity;
- PaaSapplicationsecurity;
- Customer-deployedapplication security
- IaaSapplicationsecurity
- Publiccloudsecuritylimitations

DataSecurityandStorage

- Majorityofcloudservice providersstore customers'dataonlargedatacenters.
- Although cloud service providers say that data stored is secure and safe in the cloud, customers'datamaybe damagedduringtransitionoperationsfromortothecloudstorage provider.
- Whenmultipleclientsusecloudstorageorwhenmultipledevicesaresynchronizedbyone user, data corruption may happen.
- Differentencryptiontechniqueslikepublicandprivatekeyencryptionfordatasecuritycan be used to control access to data.
- Backups or use of multiple providers can help companies to protect services from such failure and ensure data integrity in cloud storage.
- **Security** in cloud computing is a major concern. Data in cloud should be stored in encrypted form. To restrict client from accessing the shared data directly, proxy and brokerage services should be employed.

DataprivacyandsecurityIssues

- There are numerous security issues for cloud computing as it encompasses many technologies including networks, databases, operating systems, virtualization, resource scheduling, transaction management, load balancing, concurrency control and memory management.
- Therefore, security issues for many of these systems and technologies are applicable to cloud computing. For example, the network that interconnects the systems in a cloud has to be secure.
- Virtualization paradigm in cloud computing leads to several security concerns. For example, mapping the virtual machines to the physical machines has to be carried out securely.
- Datasecurityinvolves encryptingthedataaswellasensuringthatappropriatepoliciesare enforced for data sharing.
- Inaddition, resourceallocationandmemory managementalgorithmshavetobesecure.
- Thesesix areasof securityare:
  (1) securityofdata atrest,
  (2) securityofdataintransit,
  (3) authenticationofusers/applications/processes,
  (4) robustseparationbetweendatabelongingtodifferentcustomers,
  (5) cloudlegalandregulatoryissues, and
  (6) incidentresponse.



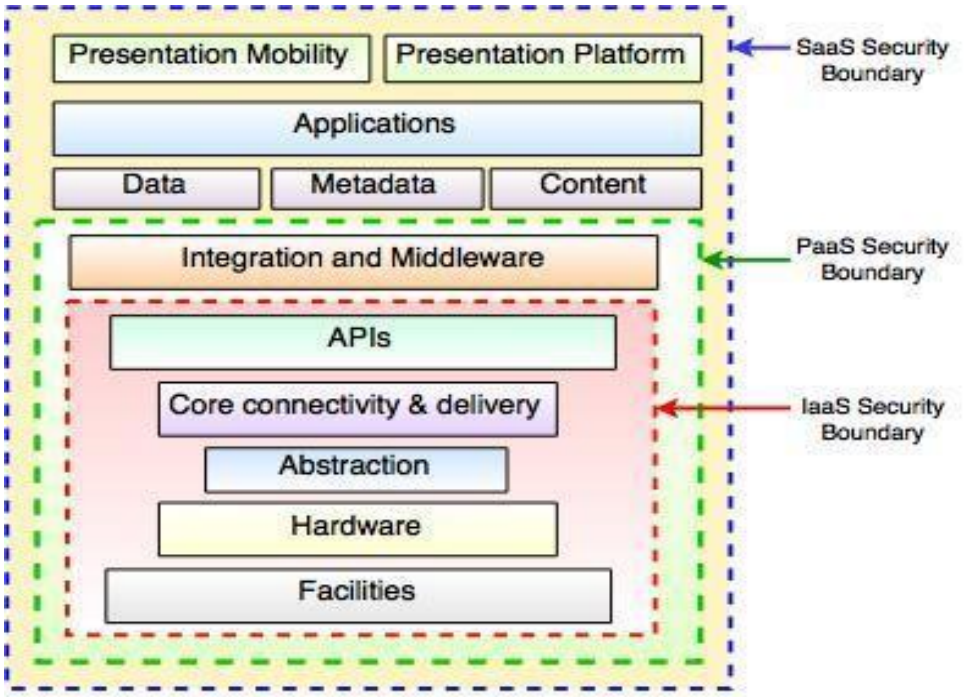
SecurityPlanning

Beforedeployingaparticularresourcetocloud,oneshouldneedtoanalyze severalaspectsof the resource such as:

- Selectresourcethatneeds tomovetothecloudand**analyzeits sensitivityto risk**.
- Considercloudservicemodelssuchas IaaS,PaaS,andSaaS.Thesemodelsrequirecustomer to be responsible for security at different levels of service.
- Considerthecloudtypetobeused suchaspublic,private,communityorhybrid.
- Understandthecloudserviceprovider'ssystemaboutdatastorageandits transferintoand out of the cloud.
- Therisk in cloud deploymentmainly depends upon theservicemodels and cloud types.

UnderstandingSecurityofCloud:SecurityBoundaries

- A particular service model defines the boundarybetween theresponsibilities of serviceprovider and customer.
- **CloudSecurityAlliance(CSA)**stackmodeldefinestheboundariesbetweeneachservice model and shows how different functional units relate to each other.
- Thefollowing diagramshows theCSA stack model:



KeyPointstoCSAModel

- IaaSisthemostbasiclevelofservicewithPaaSandSaaSnexttwoabovelevelsofservices.
- Movingupwards,eachoftheserviceinheritscapabilitiesandsecurityconcernsofthe model below.
- IaaS provides the infrastructure, PaaS provides platform development environment, andSaaS provides operating environment.
- IaaShastheleastlevelofintegratedfunctionalitiesandintegratedsecuritywhileSaaShas the most.
- Thismodeldescribesthesecurityboundariesatwhichcloudserviceprovider'sresponsibilities end and the customer's responsibilities begin.
- Any security mechanism below the security boundary must be built into the system andshould be maintained by the customer.
- Althougheachservicemodelhassecuritymechanism,thesecurityneedsalsodependupon where these services are located, in private, public, hybrid or community cloud.

UnderstandingDataSecurity

- SinceallthedataistransferredusingInternet,datasecurityisofmajorconcerninthecloud. Here are key mechanisms for protecting data.
    - AccessControl
    - Auditing
    - Authentication
    - Authorization

- All of the service models should incorporate security mechanism operating in all above-mentioned areas.

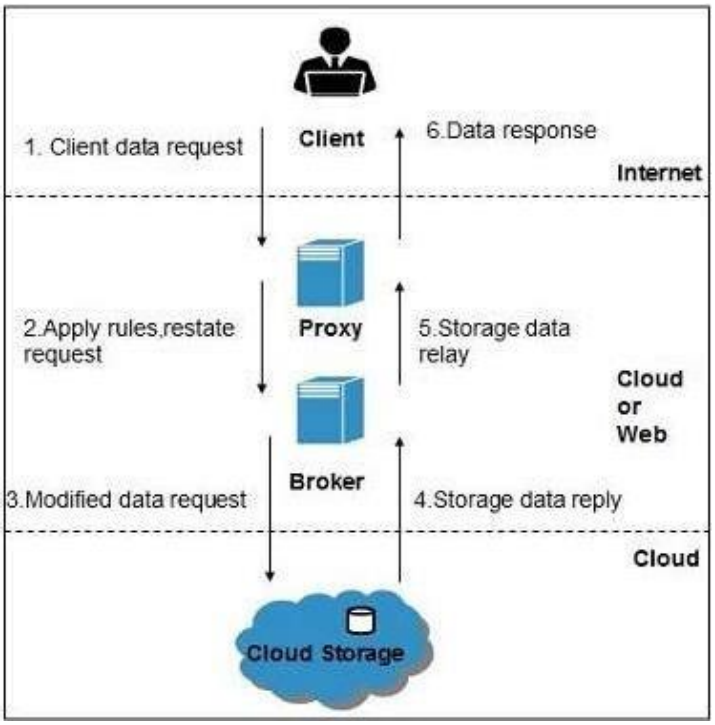
IsolatedAccesstoData

- Since data stored in cloud can be accessed from anywhere, we must have a mechanism to isolate data and protect it from client's direct access.
- Brokered Cloud Storage Accessis an approach for isolating storage in the cloud. In thisapproach, two services are created:
    - Abrokerwithfullaccesstostoragebutnoaccessto client.
    - Aproxy with noaccess to storagebut access toboth client and broker.

WorkingOfBrokeredCloudStorageAccessSystem

Whentheclientissuesrequesttoaccessdata:

- Theclient datarequest goesto theexternal serviceinterfaceof proxy.
- Theproxyforwards the request tothe broker.
- Thebroker requeststhedatafromcloudstorage system.
- Thecloudstoragesystemreturnsthedatatothe broker.
- Thebrokerreturns thedata to proxy.
- Finallytheproxysends the datato the client.
- All of theabovesteps are shown in thefollowing diagram:



Authenticationincloudcomputing

- There are multiple methods to authenticate cloud users and many issues that come along with these methods.
- Cloud computing is helping businesses to store a large amount of data at relatively low costs but it is essential these service providers offer methods to ensure users are authenticated.
- There are multiple authentication techniques in cloud computing suited for different applications and use cases when it comes to the cloud.
- Thebestcloudauthenticationmethoddependsonyourpreferencesbuteachisasupported method.

CloudAuthenticationMethods

1. **APIKeys**
   - Thismethoddoesn’trequireclientlibrariesandistransparenttotheuser.
   - Thismethodidentifiestheprojectbycreatingastrongassociationbetweenakeyanda project.
   - APIkeys arelesssecure as they arevulnerableto man-in-the-middle attacks.
   - APIkeyscaneasilybeaddedtoanyHTTPcallasaqueryparameterintheheader because they don’t require a client library.
2. **FirebaseAuthentication**
   - Thistypeofauthenticationprovidesbackendservices,appSDKs,andlibrariesto authenticate users to a mobile or web app.
   - This method authenticates users, using avariety ofcredentials likeGoogle, Facebook, Twitter or GitHub.
   - The Firebase authentication method uses a client library to sign a JSON Web Token, JWT, with a private key after the user has successfully signed in.
   - ThismethodthenvalidatestheJWT,throughaproxy,wassignedbyFirebaseandthat the issuer matches the setting in API configuration.
3. **Auth0 Authentication**
   - This method not only authenticates and authorizes apps and APIs but it is also stack, device, and identity agnostic.
   - Thismethodsupportsseveralprovidersandsecurityassertionmarkuplanguagespecification .
   - MuchlikeFirebaseAuthentication,thismethodalsoprovidesbackendservices,SDKs and user interface libraries for authenticating users in web and mobile apps.
   - Also, like Firebase Authentication, this method validates the JWT was signed and the issuer matches the API configuration.
4. **GoogleAuthentication**
   - ThisauthenticationmethodallowsuserstoauthenticatebysigninginwiththeirGoogle account.

- Oncetheuserisauthenticated,theyhaveaccesstoallGoogleservicesandaGoogleID token can be used to make calls to Google APIs and Cloud Endpoints APIs.
- ThismethodalsoverifiesthattheJWTwassignedbyGoogleandtheissuerislistedon the API configuration.

5. **GoogleAuthorization andServiceAccounts**
   - Withthismethod,aJWTcanbegeneratedandsignedusingaserviceaccountand Google-provided client library for a Google Cloud Platform project.
   - This method uses the public key to validate a Google-signed JWT and to ensure that Google is listed as the issuer in the API configuration.
   - Forthismethod, GoogleIDtokens arerecommendedforserviceaccounts becausethe API producer only needs to white list Google as an issuer for all service accounts.

CloudComputingAuthenticationIssues

1. PrivacyIssues
2. LackofTransparency
3. Security Issues
4. ThePossibility ofExploitation oftheAuthentication Mechanism
5. DifferentAuthenticationTechnologiesPresentsChallengestoCustomers

- When it comes to cloud computing, service providers require customers to store their account information in the cloud, giving service providers access to this information.
- Formanycustomers,thispresentsaprivacyissueforthem.Thelackoftransparencyinthe cloud makes it difficult for customers to ensure the proper rules are enforced.
- Customersusingmultiplecloudserviceshavemorecopiesoftheirinformationouttherein the cloud. This causes security issues for customers and cloud service providers.
- Multiple copies of accounts lead to multiple authentication processes and provide the possibility to exploit the authentication mechanism.
- Cloud service providers use different authentication technologies for authenticating users and while this has less of an impact on SaaS than PaaS and IaaS, it presents challenges to customers.
- The major importance of authentication in cloud computing is for users to ensure their projects and information are safe and there when they need it.
- While there are still a few issues associated with cloud service providers being able to performauthenticationmethodswithoutanychallengesorsecurityfears,itisimportantto remember just how new cloud computing is and the amount of room it has for progress.

CloudcontractingModel

1. **Selectingacloudservice:** Choosingtheappropriatecloudserviceanddeploymentmodel is the critical first step in procuring cloud services.
2. **Cloudserviceproviderandend-useragreements:**Termsofserviceandall CSP/customer-required agreements need to be integrated fully into cloud contracts.

3. **Service-level agreements:** SLAs need to define performance with clear terms and definitions, demonstrate how performance is being measured, and specify what enforcement mechanisms are in place to ensure that SLAs are met.
4. **CSP,agency,andintegratorrolesandresponsibilities:**Carefuldelineationbetweenthe responsibilities and relationships among the federal agency, integrators and the CSP are needed in order to effectively manage cloud services.
5. **Standards:** The use of the National Institute of Standards and Technology's Cloud ComputingReferenceArchitectureandagencyinvolvementinstandardsarenecessaryfor cloud procurements.
6. **Security:** Agencies mustclearly detail the requirements forCSPs tomaintain the security and integrity of data existing in a cloud environment.
7. **Privacy:**Ifcloudserviceshost"privacydata,"agenciesmustadequatelyidentifypotential privacy risks and responsibilities and address those needs in the contract.
8. **E-discovery:** Federal agencies must ensure that all data stored in a CSP environment is available for legal discovery by allowing all data to be located, preserved, collected, processed, reviewed and produced.
9. **Freedom of Information Act:** Federal agencies must ensure that all data stored in a CSP environment is available for appropriate handling under FOIA.
10. **E-records:** Agencies must ensure that CSPs understand and assist federal agencies in compliance with the Federal Records Act and obligations under that law.

COMMERCIALANDBUSINESSCONSIDERATIONS

Therearеanumberof contract types

Consumerto business:

- Typically these contracts relate to free cloud services, such as Facebook, where the cloud providermakesitsmoneythroughadvertisingand/orthesecondaryprocessingofcustomer data.
- Thistypeofcontracthasnoscopefornegotiation,andconsumersgenerallyhavefewrights under the contract.

Business to business:

- Thesecontractsgenerallyrelatetoserviceswhichanenterpriseispayingfor.
- There is usually little scope for negotiation, but the contract will usually vest more rights totheconsumer–althoughthecloudprovider'sliabilityforserviceperformance(including data damage and loss) may in some cases be very limited.
- Thecontractmayalsopermitthecloudprovidertounilaterallymodifyboththeserviceand thecontract,andplacetechnicalandcontractualconstraintsonswitchingfromoneprovider to another.

Bespoke(personalized) contracts:

- Cloudprovidersrarelyofferscopefornegotiationoftheircontracts,itisnotcorrecttosay that there is never any negotiation.

- Cloud providers have been known to negotiate specific agreements with those consuming organizations viewed as particularly influential or large volume.
- Cloudcontractsvary:somearebalancedandfairtobothparties,andothersareunbalanced, favoring the cloud provider.

Unit-3
Part1-PlatformasaService(PaaS)

IntroductiontoPaaS:WhatisPaaS?

- PaaScloudcomputingplatformisadeveloper programmingplatformwhichiscreatedfor the programmer to develop, test, run and manage the applications.
- Adeveloperisabletowritetheapplicationaswellasdeployitdirectlyintothislayereasily.
- PaaSextendandabstracttheIaaSlayerbyremovingthehassleofmanagingtheindividual virtual machine.
- In PaaS cloud computing platform, back end scalability is handled by the cloud serviceprovider and the end user does not have to worry about to manage the infrastructure.
- Alltheinfrastructuretorun theapplications willbe overthe internet.

AdvantagesofPaaScloudcomputinglayer

1. **Simplified Development**
   Developerscanfocusondevelopmentandinnovationwithoutworryingaboutthe infrastructure.
2. **Lowerrisk**
   Norequirementsofup-frontinvestmentinhardwareandsoftware.Developersonlyneeda PC and an internet connection to start building applications.
3. **Prebuiltbusinessfunctionality**
   Some PaaS vendors also provide already defined business functionality so that users can avoidbuildingeverythingfromveryscratchandhencecandirectlystarttheprojectsonly.
4. **Instant community**
   PaaS vendors frequently provides online communities where developer can get the ideas, share experiences and seek advice from others.
5. **Scalability**
   Applicationsdeployedcanscalefromonetothousandsofuserswithoutanychangestothe applications.

DisadvantagesofPaaScloudcomputinglayer

1. **Vendorlock-in**
   One have to write the applications according to the platform provided by PaaS vendor so migration of an application to another PaaS vendor would be a problem.
2. **Data Privacy**
   Corporatedata,whetheritcanbecriticalornot,willbeprivatesoifitisnotlocatedwithin the walls of the company there can be a risk in terms of privacy of data.
3. **Integrationwiththerest ofthesystems applications**

It may happen that some applications are local and some are in cloud. So there will be chances of increased complexity when we want to use data which in the cloud with the local data.

TopvendorswhoareprovidingPaaScloudcomputingplatform

1. GoogleApps Engine(GAE)
2. SalesFroce.com
3. WindowsAzure
4. AppFog
5. Openshift
6. CloudFoundaryfromVMware

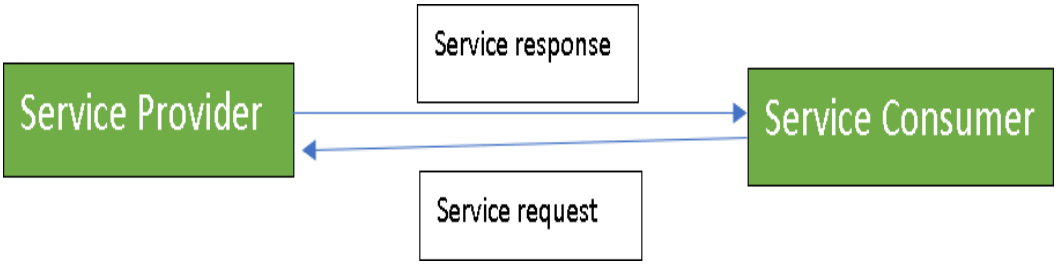
ServiceOrientedArchitecture(SOA)

Service-Oriented Architecture (SOA) is an architectural approach in which applications make use of services available in the network. In this architecture, services are provided to form applications, through a communication call over the internet.

- SOA allows users to combine a large number of facilities from existing services to form applications.
- SOA includes a set of design principles that structure system development and providemeans for integrating components into decentralized system.
- SOAbasedcomputingpackagesfunctionalitiesintoasetofinteroperableservices,which canbeintegratedintodifferentsoftwaresystemsbelongingtoseparatebusinessdomains.
- SOAprovidesaccesstoreusableWebservicesoveraTCP/IPnetwork,which makesthis an important topic to cloud computing going forward.

Therearetwomajorroleswithin Service-oriented Architecture:

- **Serviceprovider:**Theserviceprovideristhemaintaineroftheserviceandthe organization that makes available one or more services for others to use.
- **Serviceconsumer:**Theserviceconsumercanlocatetheservicemetadatainthe registry and develop the required client components to bind and use the service.



AdvantagesofSOA:

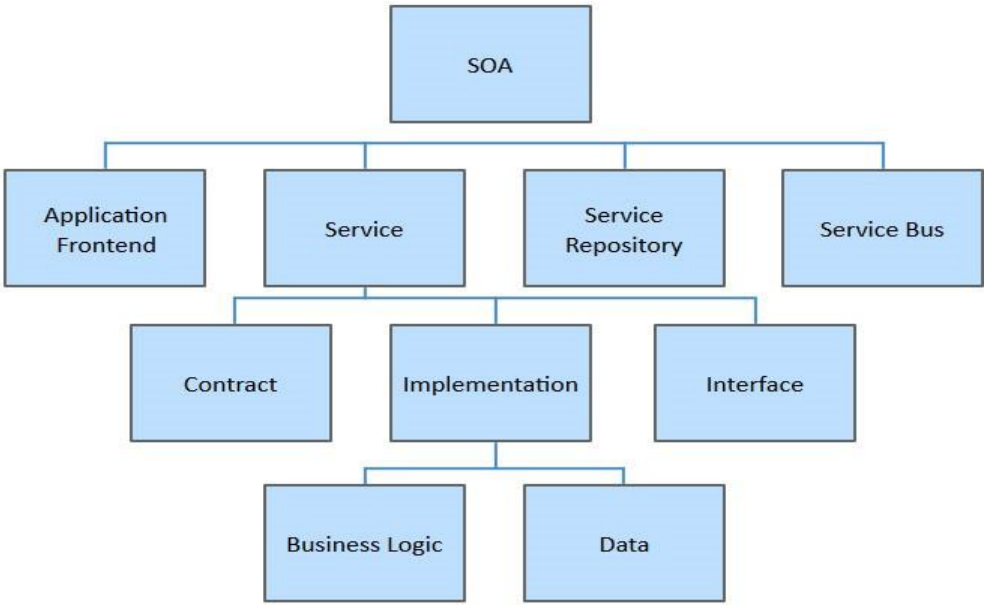
- **Servicereusability:**InSOA,applicationsaremadefromexistingservices.Thus, services can be reused to make many applications.
- **Easymaintenance:**Asservicesareindependentofeachothertheycanbeupdatedand modified easily without affecting other services.

- **Platformindependent:**SOAallowsmakingacomplexapplicationbycombining services picked from different sources, independent of the platform.
- **Availability:**SOAfacilitiesareeasilyavailabletoanyoneonrequest.
- **Reliability:**SOAapplicationsaremorereliablebecauseitiseasytodebugsmall services rather than huge codes
- **Scalability:**Servicescanrunondifferentserverswithinanenvironment,thisincreases scalability

DisadvantagesofSOA:

- **Highoverhead:**Avalidationofinputparametersofservicesisdonewheneverservices interact this decreases performance as it increases load and response time.
- **Highinvestment:**AhugeinitialinvestmentisrequiredforSOA.
- **Complex service management:** When services interact they exchange messages to tasks. the number of messages may go in millions. It becomes a unmanageable task to handle a large number of messages.

Elements/ComponentsofSOA



PracticalapplicationsofSOA:

SOAis used inmany ways aroundus whether itis mentioned or not.

1. SOAinfrastructureisusedbymanyarmiesandairforcetodeploysituationalawareness systems.
2. SOAisused toimprovethehealthcare delivery.
3. Nowadaysmanyappsaregamesandtheyuseinbuiltfunctionstorun.Forexample,an app might need GPS so it uses inbuilt GPS functions of the device. This is SOA in mobile solutions.
4. SOA helps maintain museums a virtualized storage pool for their information and content.

Unit-3
Part2-SoftwareasaService(SaaS)

IntroductiontoSaaS

- SaaS is a software distribution model in which applications are hosted by a cloud service provider and made available to customers over internet. SaaS is also known as "On-Demand Software".
- In SaaS, software and associated data are centrally hosted on the cloud server. SaaS is accessed by users using a thin client via a web browser.

AdvantagesofSaaScloudcomputinglayer

1. **Easy to buy**
   SaaS pricing is based on a monthly fee or annual fee, SaaS allows organizations to access business functionality at a low cost which is less than licensed applications.
   Unlike traditional software which is sold as a licensed based with an up-front cost (and often an optional ongoing support fee), SaaS providers generally pricing the applications using a subscription fee, most commonly a monthly or annually fee.
2. **Lesshardware required**
   The software is hosted remotely, so organizations don't need to invest in additional hardware.
3. **LowMaintenance required**
   Softwareas a service removes the necessity of installation, set-up, and often daily unkeep and maintenance for organizations. Initial set-up cost for SaaS is typically less than the enterprise software. SaaS vendors actually pricing their applications based on some usage parameters, such as number of users using the application. So SaaS does easy to monitor and automatic updates.
4. **Nospecialsoftwareorhardwareversionsrequired**
   All users will have the same version of software and typically access it through the web browser.SaaSreducesITsupportcostsbyoutsourcinghardwareandsoftwaremaintenance and support to the IaaS provider.

DisadvantagesofSaaScloudcomputinglayer

1. **Security**
   Actuallydataisstoredincloud,sosecuritymaybeanissueforsomeusers.However,cloud computing is not more secure than in-house deployment. Learn more cloud security.
2. **Latencyissue**
   Because the data and application are stored in cloud at a variable distance from the end user, so there is a possibility that there may be more latency while interacting with the applicationthanalocaldeployment.So,SaaSmodelisnotsuitableforapplicationswhose demand response times are in milliseconds.

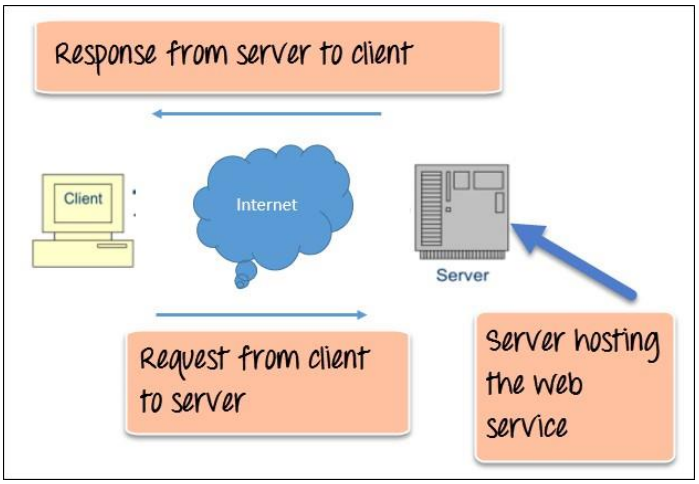
3. **TotalDependencyonInternet**
   Withoutinternetconnection,mostSaaSapplicationsarenotusable.
4. **SwitchingbetweenSaaSvendorsisdifficult**
   Switching SaaS vendors involves the difficult and slow task of transferring the very large datafilesovertheInternetandthenconvertingandimportingthemintoanotherSaaS also.

WhatisWebService?

Web service is a standardized medium to propagate communication between the client andserver applications on the World Wide Web.

Awebserviceisa softwaremodulewhichisdesigned toperformacertainsetof tasks.

- Thewebservicescanbesearchedforoverthenetworkandcanalsobeinvokedaccordingly.
- When invoked the web service would be able to provide functionality to the client which invokes that web service.



- Theclientwouldinvokeaseriesofwebservicecallsviarequeststoaserverwhichwould host the actual web service.
- These requests are made through what is known as **remote procedure calls**. Remote Procedure Calls**(RPC)** are calls made to methods which are hosted by the relevant web service.
- As an example, **Amazon provides** a web service that provides prices for products sold online via amazon.com.
- Thefrontendorpresentationlayercanbein**.NetorJava**buteitherprogramminglanguage would have the ability to communicate with the web service.
- The main component of a web service is the data which is transferred between the client and the server, and that is XML.
- **XML (Extensible markup language)** is a counterpart to HTML and easy to understand the intermediate language that is understood by many programming languages.
- SowhenapplicationstalktoeachothertheyactuallytalkinXML.

- Thisprovidesacommonplatformforapplicationdevelopedinvariousprogramming languages to talk to each other.
- WebservicesusesomethingknownasSOAP**(SimpleObjectAccessProtocol)**for sending the XML data between applications.
- Thedata is sent over normal HTTP.
- Thedatawhichis sentfromtheweb servicetotheapplication iscalledaSOAP message.
- TheSOAP messageis nothing but an XML document.
- Since the document is written in XML, the client application calling the web service can be written in any programming language.

TypeofWeb Service

Therearemainly twotypes ofweb services.

1. SOAPwebservices.
2. RESTfulwebservices.

Advantagesofwebservices

1. **Exposing Business Functionality on the network** - A web service is a unit of managed code that provides some sort of functionality to client applications or end users.
2. **Interoperability amongst applications** - Web services allow various applications to talk to each other and share data and services among themselves.
3. **AStandardizedProtocolwhicheverybodyunderstands**-Webservicesusestandardized industry protocol for the communication.
4. **Reduction in cost of communication** - Web services use SOAP over HTTP protocol, so you can use your existing low-cost internet for implementing web services.

Web 2.0

- Web 2.0 is the name used to the describe the second generation of the world wide web, where it moved static HTML pages to a more interactive and dynamic web experience.
- Web2.0isfocusedontheabilityforpeopletocollaborateandshareinformationonlinevia social media, blogging and Web-based communities.
- Web 2.0 are websites and applications that make use of user-generated content for end-users.
- Web 2.0 is characterized by greater user interactivity and collaboration, more persistent network connectivity and enhanced communication channels.
- One of the most significant differences between Web 2.0 and the traditional World Wide Web(WWW,retroactivelyreferredtoasWeb1.0)isgreatercollaborationamongInternet users, content providers and enterprises.

ElementsofWeb2.0

- **Wikis:**Websitesthatenableuserstocontribute,collaborateandeditsite content. Wikipedia is one of the oldest and best-known wiki-based sites.
- TheincreasingprevalenceofSoftwareasaService(SaaS),webappsandcloudcomputing rather than locally-installed programs and services.

- **Mobile computing**, also known as**nomadicity**, the trend toward users connecting from wherever they may be.
- **Mash-ups**:Webpagesorapplicationsthatintegratecomplementaryelementsfromtwoor more sources.
- **Social networking:** The practice of expanding the number of one's business and/or social contactsbymakingconnectionsthroughindividuals.Socialnetworkingsites **include Facebook, Twitter,LinkedIn and Google+**.
- **User-generatedcontent(UGC):**Writing,images,audioandvideocontent--amongother possibilities -- made freely available online by the individuals who create it.
- **Unified communications (UC):** The integration of multiple forms of call and multimedia/cross-media message-management functions controlled by an individual user for both business and social purposes.
- **Social curation**: The collaborative sharing of content organized around one or more particularthemesortopics.SocialcontentcurationsitesincludeReddit,Digg,Pinterestand Instagram.

Web OS

- WebOSisdesignedasa distributed system.
- TheWebOSframework enablesanewparadigmfor Internetservices.
- WebOSgoalistoprovideaplatformwhichallowstheusertobenefitfromthe computational potential offered by the Web.
- It’saimedistomakeavailabletoallsitesofthenetworkresourcestoexecutecomputations for which local resources are missing.
- The kernel of a Web OS node is a system, a reactive system responding to requests from users or system.
- AWebOS-nodeintegratesuser,server, andapplications.
- It is capable of providing a set of services, which can pass on to each other requests when appropriate.
- Each Web OS node is using its own warehouses to store and continuously update information about the node and available services and resources.
- It'sbasicallyavirtualdesktopthatgivesusercommunicationtoolslikeemail,productivity tools like word processing and ability to play games and any other application that user would find on a typical OS like Microsoft Windows.
- Web OS is network based service where a user can access his system through network. It is an introduction of one or more operating system in the category of Web operating systems.
- WebOSwasintroducedasathoughtthatonemightbeabletoplaywithapplication,store data, for sharing on the web from anywhere.
- Today Web OS are capable of storing large amount of data as large as 30 GB. User can share applications.

TaskofWebOS

- Intheseacompanyprovidescomputerservicestousersthroughan Internetconnection.
- Theproviderruns asystem ofcomputersthat includeapplication serversand databases.
- Withsomesystems,peopleaccesstheapplicationsusingWebbrowsers.
- Withothersystems, usersmust download aprogram that createsasystem-specificclient.
- Aclientissoftwarethataccesses informationorservicesfromother software.
- AWebOSmightlooklikeatraditionalOS,butitdoesn'tmanageusercomputer'shardware or software.
- AWebOSallowsusertoaccessapplicationsstorednotonusercomputer,but ontheWeb.
- TheapplicationsexistwhollyorinpartonWebserverswithinaparticularprovider network.
- When user saves information in an application, user might not store it on user computer. Instead, user saves the information to databases connected to the Internet.
- Some Web operating systems also give user the option to save information to user localhard disk drive.

CHARACTERISTICSOFWEBOPERATINGSYSTEM:

- Freesignupforpersonaluse
- Onlineapplication services.
- Nomaintainchargeandsecuritycost
- Accesstoany Website
- Workasasocial networking
- WhiletheoperatingsystemwillworkaswellastheWebsite.

FEATURESAVAILABLEINWEBOS

- **OpenAPI:**thetypologyofApplicationProgrammingInterfacethatituses
- **Open-source:** the possibility for users and developers to contribute to the enrichment of the Web OS by creating new applications and widgets
- **Integratedapps:**theapplicationsthattheWebOSfeaturesbydefault(texteditor)
- **Audio-videoPlayer:**theavailabilityofamediaplayerwithintheWebOS
- **Photoeditingtool:**theavailabilityofatoolthatallowstheusertoedit images.
- **E-mailclient:**thepresenceofanemailclient.
- **InstantMessaging:**theavailabilityofanintegratedinstantmessenger.
- **Calendar:**theexistenceofacalendar.
- **Collaboration**-Conferencingtools:thepresenceofonlinecollaborationtools(suchasVoIP)
- **Mini-Browser:**theintegrationofaWeb browser
- **Filestorage:**theamountofspaceifavailabletostorefilesonline
- **Filesharingsupport:** thesupportforfilesharingwithother users.
- **DesktopSearch:**thepresenceofadesktopsearchenginewithintheWebOS

CaseStudyonSaaS

- **Organization:British Gas**
- **SaaSprovider:Salesforce.com**
- **Application: CRM -** Customer relationship management *(*CRM) is a technology for managing all your company's relationships and interactions with customers and potential customers.
- **British Gas Services**, part of the Centrica group, is the leading domestic central heating and gas appliance installation company in the UK With **no centralized CRM system**, its Central Heating Installations (CHI) business wanted to overcome the inefficiencies. The business also wanted to improve on the manual, paper-based customer support processes that made tracking, accountability, and the ability to capture institutional knowledge difficult.
- BGS CHI implemented Salesforce.com and has seen considerable benefits. "Automated workflows route leads to appropriate individuals and provide preset responses to web-captured leads," says Sexton.
- 'speedtomarket'iscriticalforus andtherefore, as thereisno infrastructureto"standup", the solution can be ready to switch on in four months rather than six to 12 months for traditional solutions.
- Finally, cost profiling looks different as there is no large capital outlay as with a normal license purchase over time.
- However,whereSaaScostisconsistent anon-premisesolutionwillbecomecheaper.
- "We will now be able to systematically track sales leads and customer interaction, not possible with the former paper-based, disconnected support system.
- Sales and support benefit from new visibility into each other's respective activities with customerssharingonecommonviewofcustomerissuesallowsforbettercustomerservice and more insight into up-sell and cross-sell opportunities."

Unit-3
Part3-ServiceManagementinCloud

ServiceLevelAgreements (SLAs)

- A service-level agreement (SLA) is a contract between a service provider and its internal or external customers that documents what services the provider will provide and defines the service standards the provider is required to meet.
- Service providers need SLAs to help them manage customer expectations and define the situations under which they are not responsible for performance issues.
- CustomerscanalsobenefitfromSLAsinthattheydescribetheperformancecharacteristics of the service, which can be compared with other vendors' SLAs.
- Foraserviceprovider,theSLAistypicallyoneoftwofoundationalagreementsithaswith customers.
- Manyserviceprovidersestablishamasterservicesagreementtoestablishthegeneralterms and conditions in which they will work with customers.
- Earlier,incloudcomputingallServiceLevelAgreementswerenegotiatedbetweenaclient and the service consumer.
- Nowadays,withtheinitiationoflargeutility-likecloudcomputingproviders,mostService Level Agreements are standardized until a client becomes a large consumer of cloud services.
- Servicelevel agreementsarealsodefinedatdifferentlevelswhicharementionedbelow:

1. **Customer-basedSLA**
   - A customer based SLA is an agreement with one customer, covering all the services used by this customer.
   - Let'sconsiderthe relationshipbetweenyouandyourtelecomoperator.
   - You use the voice services, SMS services, data services, and several other services of the telecom operator.
   - Forallthese services,youhaveonlyonecontractbetweenyouandthetelecomoperator.
   - Similarly, if the IT service provider provides several services for the business and the customers, and if all the service levels are documented in one service level agreement for the provided services, it will be a customer based SLA.
2. **Service-basedSLA:**
   - Aservicebased SLAcoversoneserviceforall customers.
   - Let's consider that the IT service provider provides customer query service for many customers.
   - In a service based service level agreement, the service level of the customer query service will be same for all customers that will be using this service.

- For instance, if the finance department and the human resources department are two customerswhichwillbeusingthisservice,thesameSLAwillbevalidbetweentheIT service provider and these two departments since it is a service based SLA.

3. **MultilevelSLA**
   - In multi-level SLA, aspects of SLA are defined according to the organization of the customerusingsomekindofinheritancewithoveralldefinitionswithrelevanceforall subordinate levels.
   - This SLA focuses on the organization of the customer. All services and their interrelationships with subordinate services are used when defining the multi-level service level agreement structure.
   - Maintainingservicelevelagreementsarepartofservicelevelmanagement.Everytime a service change, or the service level target of a service change, the service level agreementneedstobereviewedandrevised.Thenewservicelevelagreementneedsto reflect the changes made to the service or the service level targets.

- FewServiceLevelAgreementsareenforceableas contracts,butmostlyareagreementsor contractswhicharemorealongthelinesofanOperatingLevelAgreement(OLA)andmay not have the restriction of law.
- It is fine to have an brief review the documents before making a major agreement to the cloud service provider.
- ServiceLevelAgreementsusuallyspecify**someparameters** whicharementioned below:
  1. AvailabilityoftheService (uptime)
  2. Latencyortheresponsetime
  3. Servicecomponents reliability
  4. Eachpartyaccountability
  5. Warranties

Billing&Accounting

- The billing management system mechanism is dedicated to the collection and processing of usage data as it pertains to cloud provider accounting and cloud consumer billing.
- Specifically, the billing management system relies on pay-per-use monitors to gather runtimeusagedatathatisstoredinarepositorythatthesystemcomponentsthendrawfrom for billing reporting and invoicing purposes.
- The billing management system allows for the definition of different pricing policies as well as custom pricing models on a per-cloud consumer and/or per-IT resource basis.
- Pricing models can vary from the traditional pay-per-use models to flat-rate or pay-per-allocation models, or combinations.
- Billingarrangementscanbebased onpre-usageandpost-usagepayments.
- Thelattertypecanincludepre-definedlimitsorcanbesetuptoallowforunlimitedusage.
- Whenlimits areestablished, theyareusuallyin theform ofusagequotas.

- Whenquotasareexceeded,thebillingmanagementsystemcanblockfurtherusagerequests by cloud consumers.

ComparingScalingHardware:Traditionalvs.Cloud

- Cloudcomputingisreallypopularnowadays.Moreandmorecompaniespreferusingcloud infrastructure rather than the traditional one.

Thedifferencesbetweencloudcomputingandtraditional ITinfrastructure

1. **Elasticityandflexibility**
   - Donot need tobuy thehardwareand maintain itwith yourown team.
   - Theinformationin thecloudis storedon severalservers at thesame time.
   - Itmeansthat evenif 1or2servers aredamaged,you willnot loseyourinformation.
   - Italso helpsto providethehighuptime, upto 99.9%.
   - When we talk about their traditional infrastructure, you will have to buy and maintain the hardware and equipment. If something happens, you can lose the data and spend a lot of time and money to fix the issues.
2. **Scalabilityandflexibility**
   - ThecloudcomputingistheperfectChoiceforthosewhodonotrequireahigh performance constantly but use it time by time.
   - Youcangetasubscriptionand usetheresources youpaid for.
   - Mostproviders evenlet pausethe subscriptionif youdo notneed it.
   - Atthesametime,you’reabletocontroleverythingandgetinstanthelpfromthesupport team.
   - The traditional infrastructure is not so flexible. You have to buy an equipment and maintain it even if you do not use it. In many cases, it’s even more expensive because you might need their own technical crew.
3. **Automation**
   - Oneofthebiggestdifferencesbetweencloudandtraditionalinfrastructureishowthey are maintained.
   - Cloudserviceisservedbytheprovider’ssupportteam.Theytakecareofallthe necessary aspects including security, updates, hardware, etc.
   - Thetraditionalinfrastructurerequiredtheownteamtomaintainandmonitorthe system. It requires a lot of time and efforts.
4. **Cost**
   - With cloud computing, you do not need to pay for the services you don’t use: the subscriptionmodelmeansyouchoosetheamountofspace,processingpower,andother components that you really need.
   - With traditional infrastructure, you are limited to the hardware you have. If your businessisgrowing,youwillregularlyhavetoexpandyourinfrastructure.Atthesame time, you will have to support and maintain it.

5. **Security**
   - Many people are not sure about the security of cloud services. Why can it be not so secure? As the company uses the third party solution to store data, it's reasonable to think that the provider can access the confidential data without permission. However, there are good solutions to avoid the leaks.
   - Asfortraditionalinfrastructure,youandonlyyouareresponsibleforwhowillbeable to access the stored data. Forthe companies who operate the confidential information, it's a better solution.

What kind of infrastructure is a good choice for your business? It depends on what your company does and what are your needs. Nevertheless, more and more organizations today prefer cloud infrastructure.

Economicsofscaling:Benefittingenormously

- Cloudcomputing allowsonlineaccessto acentralized datastorageandother resources.
- Thisleadstoeconomiesofscaleasmulti-tenancybasedsolutionsaredeployedononepiece of hardware, with pooled resources that are shared among different users.
- Managing cloud resources using a browser and network devices such as a smart phone,tablet and computer make operations smooth and easy.
- Animportantpartofcloudcomputingservicesisflexibilityinresourceswhichcanbe managed both up and down.
- Various companies around the world are benefiting from economies of scale that cloudcomputing & services practice brings about.
- Itoffersthe following **advantages:**
  - Fastimplementationwithloadsof timesavings
  - Fastscalingtokeeppacewithsuddenspikesin growth
  - Reductionincostofmaintaining infrastructure
  - Controloveraccess and contentas per thespecified provisioning
  - IncreaseinproductivityofITstaff
  - Reducedinvestment ininfrastructureandmaintaining facility.
  - Lookforaleadingnetworkinfrastructureprovidertomakethemostofyourinvestment in such a high-end cloud based solution.
- Youcanuse any popularwebsearchenginetolookfor one.
- Make sure you choose one that is comfortable in running all types ofapplications in the cloud and has extensive experience in handling various cloud-based projects.

ManagingData:LookingatData,Scalability&CloudServices

WhatisScalabilityinCloud?

- Scalabilityisthecapabilityofaprocess,network,softwareorappliancetogrowand manage increased demands.
- Thisis oneof themost valuableand primefeatureof cloud computing.
- Through scalability you can scale up your data storage capacity or scale it down to meet the demands of your growing business.
- Scaling in the cloud provides you the best experience of flexibility of time and money for your business.
- Whenbusiness demands areincreasing,you can easilyaddnodestoincreaseyour storage space, or you can increase the number of servers currently used.
- Whentheincreaseddemandisreducedthenyoucanmovebacktoyouroriginal configuration.
- Scalability enables you to accommodate larger workloads without disruption or complete transformation of existing infrastructure.
- Toeffectivelyforcescalabilityyouneedtounderstandthecomplexityandthetypesof scalability.
- Let'sexploredifferenttypesofscalability.

Threetypesofscalability–Vertical,HorizontalandDiagonal

1. ScaleVertically–ScaleUp:

- Vertical Scaling or Scaling up is easy, it can be done by moving the application to biggervirtualmachinesdeployedinthecloudoryoucanscaleupbyaddingexpansion units as well with your current infrastructure.
- Thisabilitytoaddresourcestoaccommodateincreasingworkloadvolumesis **vertical scaling**. It can resize your server with no change in your code.
- The downside to scaling up is that it increases storage capacity but the performance is reduced because the compute capacity remains the same.
- Workloads requiring higher throughput demand reduced latency and this can only by fulfilled by Horizontal Scaling / Scaling out.

2. ScaleHorizontally–Scaleout:

- HorizontalScalingorScalingoutistheadditionofnodestotheexistinginfrastructure to accommodate additional workload volumes.
- Contraryto Vertical Scaling, Horizontal Scaling also delivers performance along with storage capacity.
- Thetotalworkloadvolumeisaggregatedoverthetotalnumberofnodesandlatencyis effectively reduced.
- Thisscalingisidealforworkloadsthatrequirereducedlatencyandoptimized throughput.

3. ScaleDiagonally:

- Diagonalscalinghelpsyoucombinethescalingupandscaling down.
- Asthetermsuggests,scalingdownistheremovalofstorageresourcesasrequirements decrease.
- Diagonalscalingdeliversflexibilityforworkloadthatrequireadditionalstorageresources for specific instances of time.
- Forex,awebsitesetsupdiagonalscaling;asthetrafficincreases,thecompute requirements are accommodated.
- Asthetrafficdecreases,thecomputationcapacityisrestoredtoitsoriginalsize.
- Thistypeofscalingintroducesenhancedbudgetingandcosteffectivenessfor environments and businesses dealing with variable workload volumes.

ScalableCloudBasedServices:

- Infrastructure-as-a-Service(IaaS)
- Platform-as-a-Service(PaaS)
- Storage-as-a-Service(STaaS)
- Data-as-a-Service(DaaS)
- Database-as-a-Service(DBaaS)

Database&DataStoresin Cloud

- A clouddatabaseis a collection of informational content, either **structured or unstructured,** that reside on a private, public or hybrid cloud computing infrastructure platform.
- From a structural and design perspective, a cloud database is no different than one that operates on a business's own on-premises servers.
- Thecriticaldifferenceliesinwherethedatabaseresides.
- Where an on-premises database is connected to local users through a corporation'sinternallocal area network (LAN), a cloud database resides on servers and storage furnished by a cloud or database as a service (DBaaS) provider and it is accessed only through the internet.
- Toasoftwareapplication,forexample,aSQLdatabaseresidingon-premisesorinthecloud should appear identical.
- Thebehaviorofthe databaseshouldbethe same whether accessed throughdirect queries, such as SQL statements, or through API calls.
- However,itmaybepossibletodetectsmalldifferencesinresponse time.
- An on-premises database, accessed with a LAN, is likely to provide a slightly faster response than a cloud-based database, which requires a round trip on the internet for each interaction with the database.

HowCloudDatabasesWork

- Clouddatabases,liketheirtraditionalancestors,canbedividedinto twobroadcategories: **relational and non-relational.**

- Arelationaldatabase,typicallywrittenin**structuredquerylanguage(SQL),**iscomposed of a set of interrelated tables that are organized into rows and columns.
- Therelationshipbetweentablesandcolumns (fields)isspecifiedina**schema**.
- SQLdatabases,bydesign,relyondatathatishighlyconsistentinitsformat,suchas banking transactions or a telephone directory.
- Popularcloudplatformsandcloudprovidersinclude **MySQL,Oracle,IBMDB2and Microsoft SQL Server**.
- Somecloudplatformssuchas MySQLareopen sourced.
- Non-relationaldatabases,sometimes **called NoSQL,**donotemployatablemodel.
- Instead,theystorecontent,regardlessofitsstructure,asasingle document.
- Thistechnologyis**well-suitedforunstructureddata**,suchassocialmediacontent, photos and videos.

TypesofCloudDatabases

- Twoclouddatabaseenvironmentmodelsexist:traditionalanddatabaseasaservice (DBaaS).
- Inatraditionalcloudmodel,adatabaserunsonanITdepartment'sinfrastructurewith avirtualmachine.TasksofdatabaseoversightandmanagementfalluponITstaffersofthe organization.
- The DBaaS model is a fee-basedsubscription servicein which the database runs on the service provider's physical infrastructure. Different service levels are usually available. In a classic DBaaS arrangement, the provider maintains the physical infrastructure and database, leaving the customer to manage the database's contents and operation.
- Alternatively,acustomercansetupamanagedhostingarrangement,inwhichtheprovider handles database maintenance and management. This latter option may be especially attractive to small businesses that have database needs but lack adequate IT expertise.

Clouddatabasebenefits

- Compared with operating a traditional database on an on-site physical server and storage architecture, a cloud database offers the following distinct advantages:
- **Elimination of physical infrastructure**-In a cloud database environment, the cloudcomputingprovider of servers, storage and other infrastructure is responsible for maintenance and keeping high availability. The organization that owns and operates the database is only responsible for supporting and maintaining the database software and its contents. In a DBaaS environment, the service provider is responsible for managing and operating the database software, leaving the DBaaS users responsible only for their own data.
- **Costsavings**-Throughtheeliminationofaphysicalinfrastructureownedandoperatedby an IT department, significant savings can be achieved from reduced capital expenditures, less staff,decreased electrical andHVAC operating costs and asmalleramount ofneeded physical space.

- DBaaS benefits also include instantaneousscalability, performance guarantees,failover support, declining pricing and specialized expertise.
- Migratinglegacydatabasestothe cloud
- An on-premises database can migrate to a cloud implementation. Numerous reasons exist for doing this, including the following:
- AllowsITtoretireon-premisesphysical serverandstorageinfrastructure.
- Fillsthetalentgap whenITlacks adequatein-housedatabaseexpertise.
- Improvesprocessingefficiency,especiallywhenapplicationsandanalyticsthataccessthe data also reside in the cloud.
- Achievescostsavingsthroughseveralmeans,including:
- Reductionofin-houseIT staff.
- Continuallydecliningcloudservice pricing.
- Payingforonlytheresourcesconsumed,knownaspay-as-you-gopricing.

Relocating a database to the cloud can be an effective way to further enable business application performance as part of a wider software-as-a-servicedeployment. Doing so simplifies the processes required to make information available through internet-based connections.Storage consolidationcan also be a benefit of moving a company's databases to the cloud. Databases in multiple departments of a large company, for example, can be combined in the cloud into a single hosted database management system.

LargeScaleDataAnalysis

- Large scale data analysis is a broad term that encompasses a series of different tools and systems to process big data.
- Typically, large scale data analysis is performed through two popular techniques: parallel database management systems (DBMS) or MapReduce powered systems.
- TheparallelDBMSsystemrequiresthatthedatabeinaDBMSsupportedschema,whereas the MapReduce option supports data in any form.
- Moreover, the data extracted or analyzed in large-scale data analysis can be displayed in various different forms, such as tables, graphs, figures and statistical analysis, depending on the analysis system.

CASE STUDY

CaseStudyonOpenSource&CommercialClouds

Eucalyptus

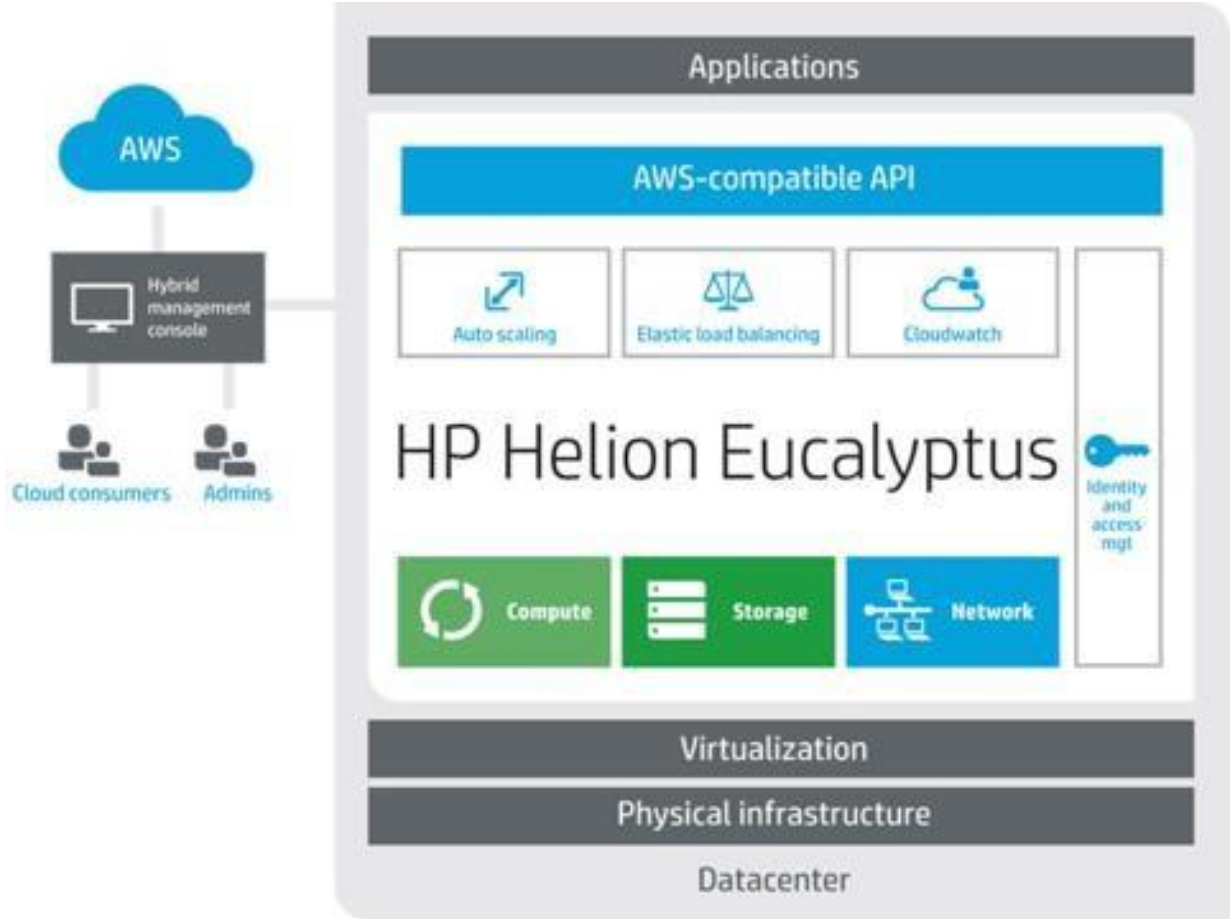
- Eucalyptus is a paid and open-source computer software for building Amazon Web Services(AWS)-compatibleprivateandhybrid cloud computing environments, originally developed by the company Eucalyptus Systems.
- Eucalyptusisanacronymfor**ElasticUtilityComputingArchitectureforLinkingYour Programs to Useful Systems**.
- Eucalyptus enables pooling compute, storage, and network resources that can be dynamically scaled up or down as application workloads change.
- MårtenMickos wastheCEOof Eucalyptus.
- InSeptember2014,Eucalyptuswasacquiredby**Hewlett-Packard**andthenmaintainedby DXC Technology.
- AfterDXCstoppeddevelopingtheproductinlate2017,AppScaleSystemsforkedthecode and started supporting Eucalyptus customers.
- EucalyptuscommandscanmanageeitherAmazonorEucalyptusinstances.Userscanalso move instances between a Eucalyptus private cloud and the Amazon Elastic ComputeCloudtocreateahybridcloud.Hardwarevirtualizationisolatesapplicationsfromcomp uter hardware details.

Eucalyptususestheterminology:

- **Images**–Animageisafixedcollectionofsoftwaremodules,systemsoftware,application software, and configuration information that is started from a known baseline (immutable/fixed). When bundled and uploaded to the Eucalyptus cloud, this becomes a Eucalyptus machine image (EMI).
- **Instances** – When an image is put to use, it is called an instance. The configuration is executedatruntime,andtheCloudControllerdecideswheretheimagewillrun,andstorage and networking is attached to meet resource needs.
- **IP addressing** – Eucalyptus instances can have public and private IP addresses. An IP addressisassignedtoaninstancewhentheinstanceiscreatedfromanimage.Forinstances that require a persistent IP address, such as a web-server, Eucalyptus supplies elastic IP addresses. These are pre-allocated by the Eucalyptus cloud and can be reassigned to a running instance.
- **Security** – TCP/IP security groups share a common set of firewall rules. This is a mechanismtofirewalloffaninstanceusing IPaddressandportblock/allowfunctionality. Instances are isolated at TCP/IP layer 2. If this were not present, a user could manipulate

the networking of instances and gain access to neighboring instances violating the basic cloud tenet of instance isolation and separation.

- **Networking**–Therearethreenetworkingmodes.InManagedMode,Eucalyptusmanages alocalnetworkofinstances,includingsecuritygroupsandIPaddresses.InSystemMode, Eucalyptus assigns a MAC address and attaches the instance's network interface to the physicalnetworkthroughtheNodeController'sbridge.SystemModedoesnotofferelastic IP addresses, security groups, or VM isolation. In Static Mode, Eucalyptus assigns IP addresses to instances. Static Mode does not offer elastic IPs, security groups, or VM isolation.
- **Access Control** – A user of Eucalyptus is assigned an identity, and identities can be grouped together for access control.



MicrosoftAzure

WhatisAzure?

- Azure is Microsoft's cloud platform, just like Google has it's Google Cloud and Amazon has it's Amazon Web Service or AWS.000.
- Generally,itisa platformthroughwhichwecanuseMicrosoft'sresource.
- Forexample,tosetupahugeserver,wewillrequirehugeinvestment,effort,physicalspace and so on.

- In such situations, Microsoft Azure comes to our rescue. It will provide us with virtual machines, fast processing of data, analytical and monitoring tools and so on to make our work simpler.
- Thepricing ofAzureisalso simplerand cost-effective.
- Popularlytermedas“PayAsYouGo”,whichmeanshowmuchyouuse,payonlyforthat.
- MicrosoftunveiledWindowsAzureinearlyOctober2008butitwenttoliveafterFebruary 2010.
- Laterin2014,Microsoftchangeditsnamefrom**WindowsAzuretoMicrosoftAzure**.
- Azureprovidedaserviceplatformfor.NETservices,SQLServices,andmanyLive Services.
- Manypeoplewerestillveryskepticalabout“thecloud”.
- MicrosoftAzureisgetting biggerand betterincoming days.
- Moretoolsandmorefunctionalities aregettingadded.Ithas tworeleasesasof now.
- It’sfamousversion**MicrosoftAzurev1**andlater **MicrosoftAzurev2**.
- Microsoft Azurev1 was morelikeJSONscript driven thenthenewversion v2, which has interactive UI for simplification and easy learning.
- MicrosoftAzurev2is stillinthepreviewversion.

AmazonEC2

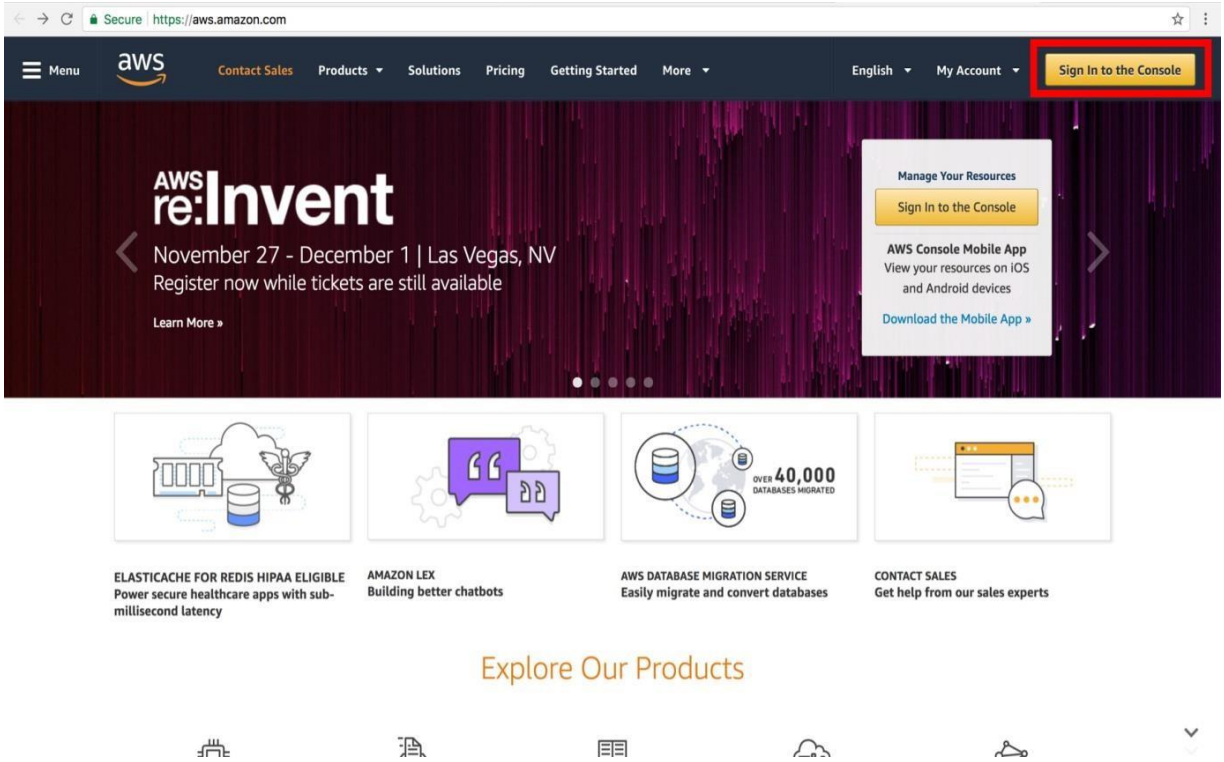
- Therearetimeswhen oneis limited bythe capabilitiesof adesktop or laptop.
- Supposeadata scientisthasalargedataset thattheywould liketodo someanalysis on.
- The scientist proceeds to try and load the entire dataset into memory and an error like the one below occurs.

```
> data <- read.csv(file = 'massiveFile.csv')
Error: cannot allocate vector of size 500.0 Mb
```

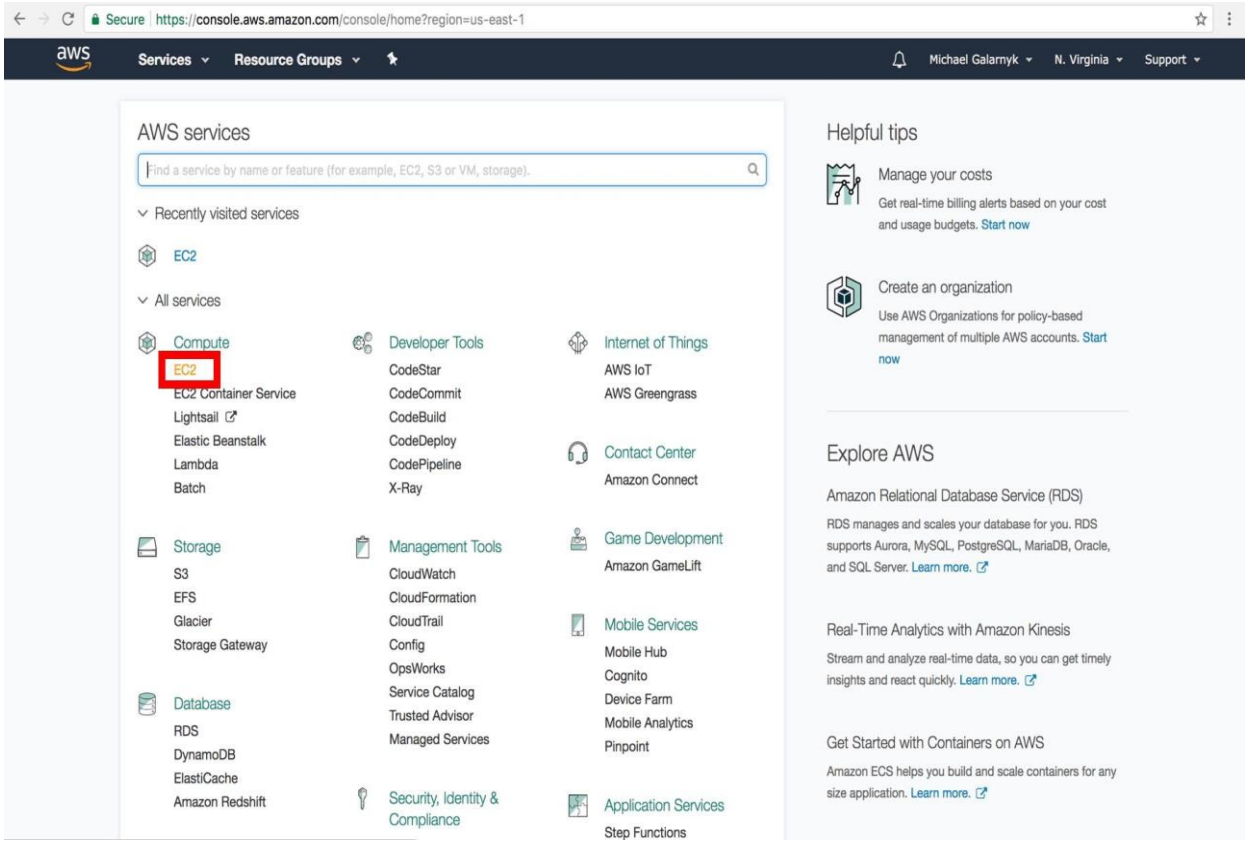
- TheerrorresultedbecausetheavailableRAMwasexhausted.
- Theoperating systemcouldn't allocateanother500Mbof RAM.
- Whiletherearemanydifferentsolutionstothistypeofproblem,onepossiblesolutioncould be to upgrade the RAM of the computer.
- Besides having to make an investment in more RAM, there are limits to how far some computerscanbeupgraded.Thepotentialsolutionexploredinthistutorialistouseavirtual machine in the cloud (AWS) with more RAM and CPU.
- VirtualmachinesonAWSEC2, alsocalledinstances,havemany advantages.
- A few of the advantages include being highly scalable, they are easy to start and stop and they allow for the selection of different platforms (operating systems).
- Animportantpointthingtoemphasizeisthatalthoughthistutorialcovershowtolauncha Windows based virtual machine, there are many different types of virtual machines for many different purposes.
- Withthat, let'sget started.

Createan AWS Account and Signinto AWS.

1. OntheAmazonWeb Servicessiteclick on"SignIn totheConsole".
2. Signinif youhaveaccount. Ifyoudon't, youwillneed tomakeone.

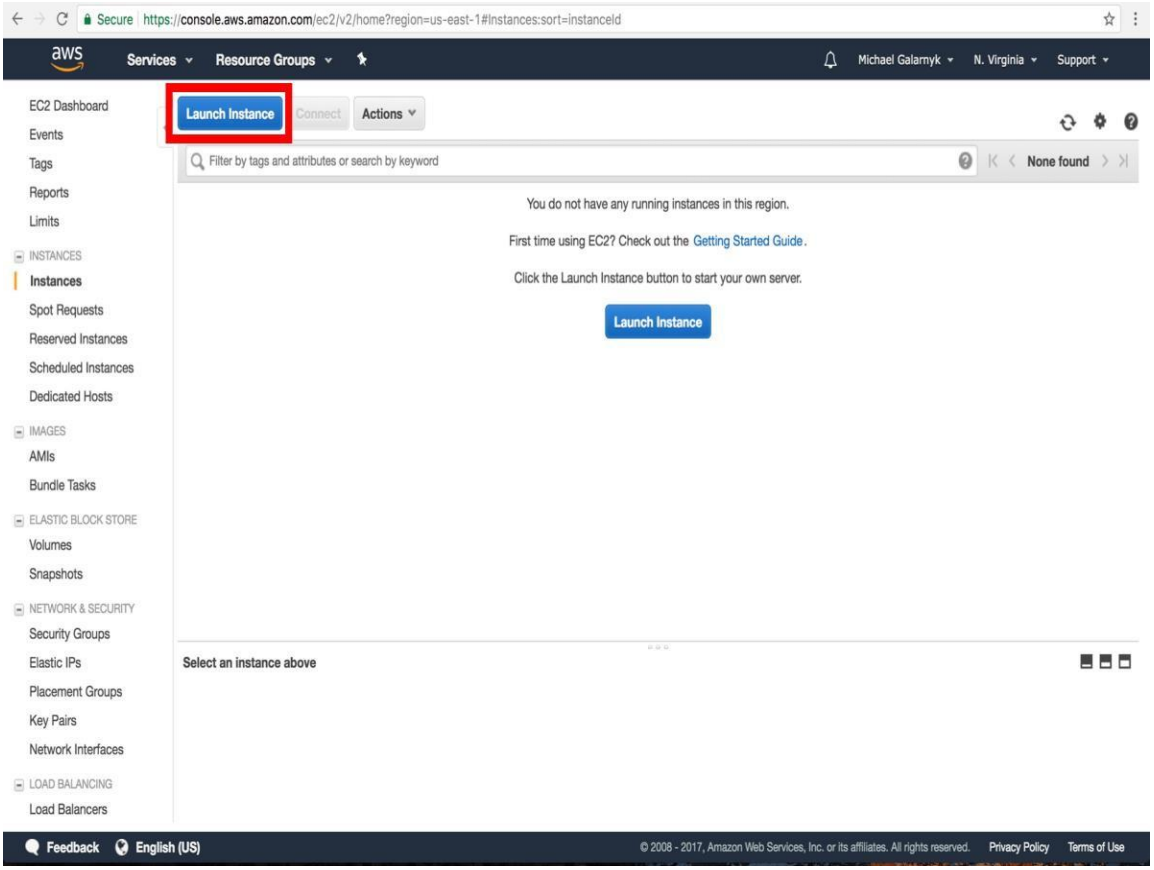


3. OntheEC2 Dashboard, clickon EC2.



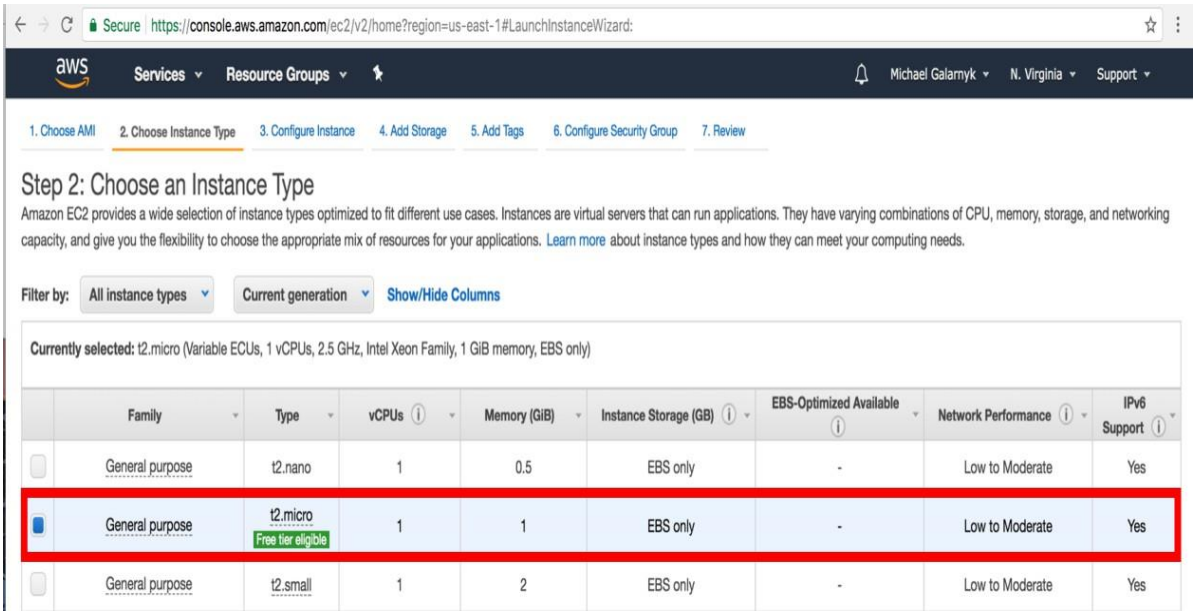
Createan Instance

4. OntheAmazonEC2console,clickonLaunchInstance.

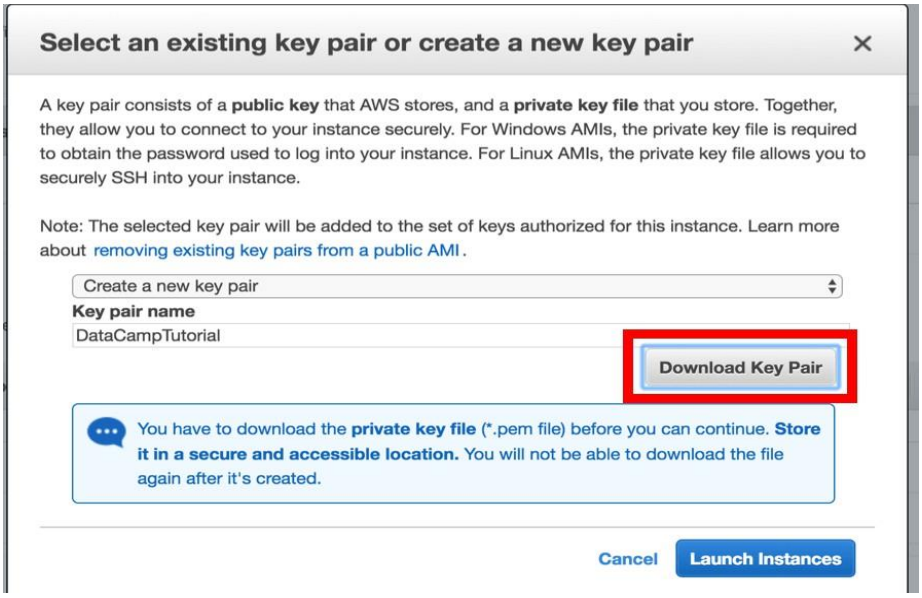


5. Clickonthe"Select"buttonintherowwithMicrosoftWindowsServer2016Base.Please note that this will create a Windows based instance instead of a typical Linux based instance. This effects how you will connect to the instance.

6. Makesuret2micro(freeinstancetype)isselected.and clickon"Reviewand Launch"

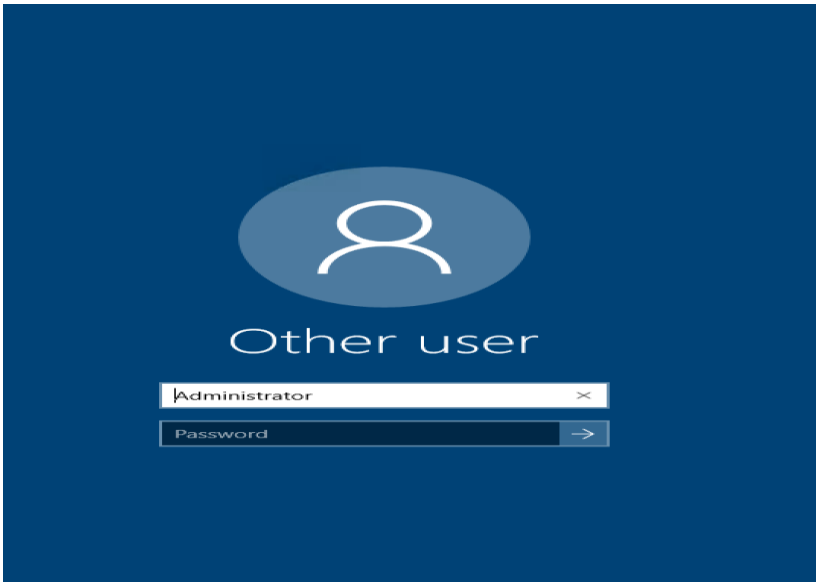


7. Clickon Launch.
8. Select"Createanewkeypair".Intheboxbelow("Keypairname"),fillinakeypairname. I named my key DataCampTutorial, but you can name it whatever you like. Click on "Download Key Pair". This will download the key. Keep it somewhere safe.
9. Next,clickon"LaunchInstances"



10. The instance is now launched. Go back to the Amazon EC2 console. I would recommend that you click on what is enclosed in the red rectangle as it will bring you back to the console.
11. Wait till you see that "Instance State" is running before you proceed to the next step. This can take a few minutes.
12. Clickon connect.
13. Click on "Download Remote Desktop File". Save the remote desktop file (rdp) file somewhere safe.

14. Click on "Get Password". Keep in mind that you have to wait at least 4 minutes after you launch an instance before trying to retrieve your password.
15. Choosethepem fileyou downloaded fromstep 7andthen click"DecryptPassword"
16. Afteryoudecryptyourpassword,saveitsomewheresafe.Youwillneedittologintoyour instance.
17. Open your rdp file. Click on continue. If your local computer is a Mac, you will need todownload"MicrosoftRemoteDesktop"fromtheAppStoretobeabletoopenyourrdpfile.
18. Enteryourpasswordyougotfromstep 15



DifferencebetweenAWS(AmazonWebServices),GoogleCloudandAzure

	AWS	Google Cloud	Azure
Technology	EC2 (Elastic Compute Cloud)	Google Compute Engine (GCE)	VHD (Virtual Hard Disk)
Databases Supported	AWS fully supports relational and NoSQL databases and Big Data.	Technologies pioneered by Google, like Big Query, Big Table, and Hadoop, are naturally fully supported.	Azure supports both relational and NoSQL databases, and Big Data, through Windows Azure Table and HDInsight.
Pricing	Per hour – rounded up	Per minute – rounded up (minimum 10 minutes)	Per minute – rounded up commitments (pre-paid or monthly)
Models	On demand, reserved, spot	On demand – sustained use	On demand – short term commitments (pre-paid or monthly)
Difficulties	Many enterprises find it difficult to understand the company's cost structure	Fewer features and services.	Less "enterprise-ready"
Storage Services	• Simple Storage Service (S3) • Elastic Block Storage (EBS) • Elastic Block Storage (EBS)	• Blob Storage • Queue Storage • File Storage • Disk Storage • Data Lake Store	• Cloud Storage • Persistent Disk • Transfer Appliance
Machine Learning	• Sage Maker • Lex • Polly • And many more	• Machine Learning • Azure Bot Service • Cognitive Service	• Cloud Speech API • Cloud Video Intelligence • Cloud Machine Learning Engine • And many more.

CasestudyonEucalyptus

- **Rafter is an educational software company** that operates BookRenter.com, which was the first online textbook rental service.
- The company's Rails-based platform requires considerable cloud resources for application development and testing. , it needed a way to lower its own spending on publiccloud infrastructureand meet developerrequirements foradditional testing and development.
- By building a Eucalyptus cloud, Rafter realized immediate advantages over its previously Amazon Web Services (AWS)-only infrastructure.
- This provided betterautomation and imagemanagement than virtualization tools such asVMwareorhostedservicessuchasRackspaceandmuchmoreeconomicallyfeasible than Amazon Reserved Instances.
- Overall,EucalyptusgavethemvariableAWScostsandbenefitfromamorepredictable and affordable configuration.
- Organizations can use or reuse AWS-compatible scripts to manage their own on-premise IaaS environments.
- The AWS API is implemented on top of Eucalyptus, so tools in the cloud ecosystem that can communicate with AWS can use the same API with Eucalyptus.
- If your company started with an entirely AWS environment, then you cango on using thesametools,scriptsandimageswithEucalyptus.Or,ifEucalyptusisthefirstpartof your cloud journey, then our deep AWS API compatibility sets you up to efficiently scale and automate hybrid cloud infrastructure as operations grow."
- EucalyptusiscompatiblewiththefollowingAWS features:
    - AmazonElastic ComputeCloud (EC2)
    - AmazonElasticBlockStorage (EBS)
    - AmazonMachineImage (AMI)
    - AmazonSimpleStorageService(S3)
    - Amazon IdentityandAccessManagement(IAM)
    - Auto Scaling
    - ElasticLoadBalancing
    - Amazon CloudWatch
- DatacenterITandDevOpsteamsareresponsibleformanagingthecompany's Eucalyptus implementation.
- EucalyptusfeaturedtheAWScompatibility,flexibilityandadvancedfeatures necessary to improve Rafter's development and test environment.
- EucalyptusworkswithpopularAWStoolsformakingiteasytomigrateapplications from the public cloud without rewriting or duplicating code.

BenefitsofEucalyptus

- Empoweredsoftwareengineerstoeasilyprovisiontheirowninstances,reducingthe IT department's burden
- EasilyintegratedwithexistingEC2-basedcodeanddev/testinfrastructure,dueto Eucalyptus' AWS API compatibility
- Allowedfor easycapacity expansionas theprivateclouddemandcontinuesto grow
- Multipliedengineeringproductivitybyusingthelatesthardwareavailable,ata fraction of the cost to similar public cloud options
- Greatlyreducedcosts,whilestillmakingiteasy toleveragehybridcapabilitiesand scale to the public cloud when needed
- Theprivate andhybridsolutionloweredcosts.
- AllowedRaftertousemuchmorepowerfulhardwarebyusingprivatecloudrather than public.
- RaftercouldretainAWS'sconsiderable capacityandelasticityforspecificworkloads.
- EucalyptusdoubledRafter'scloudcapacity,increasedengineeringproductivityand generated substantial cost savings.
- IthastwiceasmanyinstancesonEucalyptusasitdidundertheoriginalAWSsetup, with minimal monthly operating expenses. At the same time, Rafter's integration of Eucalyptus increased its flexibility with cloud vendors, preventing it from being too dependent on any single service.
- OnthecostSide,RafterhasbeenabletobettercontroltheexpensesofAmazon Reserved Instances, on-demand instances and EBS storage/snapshots.
- Eucalyptushasreplacedthehighvariable costsof AWSwithmuchlowerfixedcosts.

CasestudyonMicrosoftAzure

- Globally recognized for its ground-breaking approach towards delivering business value,increasingspeeddigitalinnovationandenablingbusinessplatformstobefuture-ready enterprises, Wipro Ltd. is one of the most successful IT companies in India and around the world.
- It is a trusted IT partner of choice for global businesses looking to 'differentiate at the front' and 'standardize at the core' through technology interventions.

WiproBeforeCloud

- Wiprobeganthetransformationofitsinternalapplicationbackgroundacoupleofyears back and has been progressing forcefully to provide a great digital experience to its employees, simplified workflows and systems, enhanced automation and innovation.

- Itsfocushasbeentodevelopapplicationsandservices,whichwouldgiveemployeesa great user experience, open up new revenue generation streams, enhance productivity and make IT more responsive and cost effective.
- To build a strong digital core, it was important that the organization transformed its legacyenterpriseapplicationsintofullyrenovated,mobileenabledandanalyticsdriven ones.
- In order to do so and cope up with the expectation of rapid app development and deployment,itwasveryimportantthattheorganizationupgradeditsinfrastructureand moved from on premises to cloud.
- Apart from elasticity in infrastructure, cloud adoption would also mean increased efficienciesrelatedtodatacenterfootprint,networkplanningandutilization,hardware refreshes,standardizedbusinessprocesses,whichwouldbringsignificantcostbenefits on a large scale.
- Additionally, since Wipro has been a pioneer in cloud adoption, it would also help in opening new streams of revenue for the organization.
- **Wiprowasthereforelookingoutforacost-effective,elasticandenterprisefriendly IT infrastructure solution.**

IntroductionofMicrosoftAzure

- PostdetailedevaluationofdifferentvendorsWiprochoseMicrosoftAzureasitCloud partnerbecauseofagility,stability,costeffectivenessandthedeepintegrationitoffered with Office Suite of apps.
- To start off with it was important for the company to have a higher availability with infrastructure, scale up and down as per demand and this aspect was fundamental for the Wipro.
- Realvalueofcloudcomeswiththeeaseofmanagement,insightsandthemicroservices
- WearehappywechosetheAzureplatformfromMicrosoft.Theyhaveworkedwithus in shaping up our platform roadmap," says Senior Vice President of Wipro Limited.

HowMicrosoftAzureBenefitedWipro

- Migrating to Microsoft Azure has given Wipro elasticity on its infrastructure, as well as increased efficiencies related to data center.
- Toreducecostofinfrastructureandtakeadvantageoftheplatform,theapplication was developed and deployed in MS Azure PaaS.
- Allthemodulesinthesystemwerewritteninanewarchitecture,WebandAPI layers.
- ThefrontendwasdevelopedusingmodernJQuerylibrariestomaketheapplication fast.
- Entireapplicationwasdevelopedanddeployedinjust6weeks.Itwasestimatedthat the overall cost of this application was reduced by more than 60%

Otherbenefitsare:

Scalability:

- ManyenterpriseapplicationsatWiprohaveacyclicalusageperiod,requiring scalability, concurrency and high performance for short period of time.
- Highavailabilityisamustduringthatperiodanddemandshighscalabilityofresources.
- Wipro can now easily scale up and down based on demand and pay for what is being consumed.

Cost-effectiveness:

- With Azure, The infrastructure costs on the cloud platform were calculated based on usage,itgavetheorganizationtheabilitytopayforwhat'sbeingusedandnotbebound to pay for unused services and resources.

Time-to-marketofnew applications:

- Earlier, due to the unavailability of the latest hardware or very time-intensive cycles, the time-to-market of the applications was much high.
- With Microsoft Azure, all these issues were solved. Microsoft acted as an end-to-end 'cloud partner' capable of providing an entire scope of services on the cloud.

Conclusion

- The Azure platform has helped us build modern applications, which offer simplicity and great user experience with extensive leverage of open source technology. And thankstothegreatsetofmicroservices,whichcouldbeconsumedwithease,theapps could be developed at a very short time span. --- **General Manager and Head of Applications, Information Systems**

IntroductiontoAWS

Amazon Web Services (AWS) is a collection of remote computing services, also called web services, that make up a cloud computing platform by Amazon.com. The mostcentralandwell-knownoftheseservicesareAmazonEC2andAmazonS3.The service is advertised as providing a large computing capacity (potentially many servers) much faster and cheaper than building a physical server farm.

Benefits:-

1. **Easytouse:-**AWSisdesignedtoallowapplicationproviders,ISVs,andvendors toquicklyandsecurelyhostyourapplications –whetheranexistingapplication or a new SaaS-based application. You can use the AWS Management Console or well-documented web services APIs to access AWS's application hosting platform.
2. **Flexible:-** AWS enables you to select the operating system, programming language,webapplicationplatform,database,andotherservicesyouneed.With AWS, you receive a virtual environment that lets you load the software and

servicesyourapplicationrequires.Thiseasesthemigrationprocessforexisting applications while preserving options for building new solutions.

3. **Cost effective:-** You pay only for the compute power, storage, and other resourcesyouuse,withnolong-termcontractsorup-frontcommitments.For more information on comparing the costs of other hosting alternatives with AWS, see the AWS Economics Center.
4. **Reliable:-** With AWS, you take advantage of a scalable, reliable, and secure globalcomputinginfrastructure,thevirtualbackboneofAmazon.com'smulti-billion dollar online business that has been honed for over a decade.
5. **Scalable and High Performance:-** Using AWS tools, Auto Scaling, and Elastic LoadBalancing,yourapplicationcanscaleupordownbasedondemand.Backed by Amazon's massive infrastructure, you have access to compute and storage resources when you need them.
6. **Secure:-** AWS utilizes an end-to-end approach to secure and harden our infrastructure,includingphysical,operational,andsoftwaremeasures.Formore information, see the AWS Security Center.

ApplicationSolutions:-AWSoffersareliableandflexiblecloudinfrastructureplatform that enables customers to run any type of business application, from small departmental solutions to mission-critical applications in a secure and robust environment.

AMAZONEC2CASESTUDY

- AmazonElasticComputeCloud(EC2)isacentralpartofAmazon.com'scloud computing platform, Amazon Web Services (AWS).
- EC2allowsuserstorentvirtualcomputersonwhichtoruntheirowncomputer applications.
- EC2 allows scalabledeployment of applications by providing a WebservicethroughwhichausercanbootanAmazonMachineImagetocreate avirtualmachine,whichAmazoncallsan"instance",containinganysoftware desired.
- Ausercancreate,launch,andterminateserverinstancesasneeded,payingby the hour for active servers, hence the term "elastic".
- EC2providesuserswithcontroloverthegeographicallocationofinstancesthat allows for latency optimization and high levels of redundancy.

Casestudy

- HDM:-Thecompanyfirstlaunchedin1999as"Hungama.com",apromotionalmarketing portal.

- In2000thecompanyacquiredIndiafmandinthefollowingyearsbegantowork marketing campaigns for companies such as Coca-Cola and Nike.
- In2007Hungamalaunchedtheirgamingportalandin2009,thecompanyre-launched theirwebsiteandcompanyname,changingittoHungamaDigitalMediaEntertainment.
- In2012HungamaDigitalMediaEntertainmentlaunchedArtistaloud!,adigitalplatform for artists and music fans.

Challenges:-

- BeforemigratingtoAWS,thecompanyranitsserversinlocaldatacenters.This solution soon became expensive, time-consuming, and inefficient.
- Hungama wanted to turn projects around quickly but old equipment caused delays in implementation and service launches.
- In2008,HungamamigratedtoAWStotakeadvantageofthecosteffectiveness, flexibility, and fast time-to-market offered by the cloud.
- "OurmigrationtoAWSin2008reducedITcosts considerably," Hungama.
- ThecostsavingsallowedHungamatofocusitsengineeringresourcesongetting products and services to market more quickly.
- Thecompanybegantogrowrapidly,butasmoreinternalteamsstartedusingAWS,its monthly costs also grew.
- Hungama'sinfrastructureteamengagedAWSSupporttohelpthemfindwaysto optimize costs.

Solutionby AWS:-

- Asacontentproviderandaggregator,Hungamarequiresenormousamountsofstorage. ThecompanyusesAmazonSimpleStorageService(AmazonS3)tohostmorethan60 TB of content.
- For server and storage management, Hungama uses Amazon Elastic Compute Cloud (AmazonEC2)andAmazonRelationalDatabaseService(AmazonRDS)withAmazon S3.TheflexibilityoftheAWSsolutionenablestheBollywoodmoviemakertodevelop applicationsonmultipleplatformsandprogrammingmodelsincludingJava,PHP, .NET,Oracle, MySQL,and SQLtechnologies.
- As anAWS Support, Enterprise-level tier customer, Hungama has access to AWS TrustedAdvisor,whichcustomerscanusetoaudittheirAWSusageagainstknownbest practices.
- AWS Trusted Advisor identifies opportunities to save money, improve system performance, and security.
- Hungama used AWS Trusted Advisor to run Cost Optimizing checks, which audited Amazon EC2 instances andAmazon Elastic Block Store(Amazon EBS) volumes in Hungama's environment.

Explaindataprivacyandsecurityissues.

CLOUDSECURITY ISSUES

- Securityisamajorissue in ITbusinessenvironment.
- SincecustomersandusersshiftedfromGridcomputingtoCloudcomputingintheir business, many security issues appeared, which is a major concern for the Cloud provider due to the risk of losing customers.
- TheCloudComputingismainlybuiltonvirtualizationenvironment,thatincrease more risk of securing the cloud.

VirtualSecurityissues:

- Thevirtualenvironmentareaofthecloudcomputingisthemostsensitiveand important part of the cloud.
- This is because all the devices in the cloud are connected virtually through virtual networksthatarerunningandmanagingtheITinfrastructuresandvirtualserversin the cloud.
- Invirtualizationtechnology,multipleVirtualMachines(VM's)canrunontopofa single physical machine, and can run on any OS within each VM's to manage the infrastructure.
- Oneofthemain**virtualsecurityissues**inthecloudareattacksonthenetwork between VM's, and the trust between different VM's .
- Thereareothersproblems,suchasnon-secureAppsandvulnerabilityinVM's,which allow any unauthorized access.

1) NetworkAttacksinVirtualcloud:

- Runningmanydifferentvirtualizationproductsincreasestheattackers(especially the hackers) perimeter.
- Amazon's cloud computing service (EC2), could be used to hack into other systemsbyusingEC2cloudservicetoallowabruteforceattack,thatwillfire 400,000 passwords per second at a secured wireless network.
- Withinaperiod oftwenty minutesthesystemwould hadbeen attacked.
- TheattackershackedandshutdownSony'sonlinecustomernetworksinApril 2011.

- Hackersusedcloud basedattacksto disruptserviceto roughly100 million
- usersworldwide.

2) DistributedDenialofService(DDOS)Attacks:

- Thisattacktargets thenetworksand servers.
- Itmakesthenetworktrafficandusersbeingdeniedtoaccessacertain Internet- based service in the cloud.
- Inworstcasestheattackerswillusebonnetstoperform DDOS.Inordertostop hackers of attacking the network, face blackmail is provided.
- DDOSattacksshouldbeconsideredasthreatsforcloudproviderssuchAWS, Google Apps, and Microsoft Cloud.
- Thesescenariosshowusthatcloudcomputingnetworkisstillnotsecure,andthis will drive us to non-secure applications.

3) Non-SecureApps:

- Cloudapplicationssecurityisacomplicatedissuefororganizationsandcustomers if they ignore securing their data before deploying it in the cloud.
- Theyneedto considerthenew threatsandattacksspread
- Non-Secure applications opens the doors for further threats that could result in attackingthecloudthroughthenetworkandApplicationProgrammingInterface (API).
- Manin TheMiddle attack is oneof theproblems for non-secure apps.
- This attack works as eavesdropping. Here, the attacker creates independent connectionswiththevictimsandtransmitsmessagesbetweenthemtomakethem believethattheyaretalkingdirectlytoeachotheroveraprivateconnectionwhen in fact the entire conversation is run and controlled by the attacker.
- FacebookapplicationispronetoMan-In-The-Middleattack(MITM)onusers' data.

4) DomainNameServer(DNS)Attacks:

- It'seasyforattackerstoattackDNSincloudcomputingwhentheusersor customers try to call the server by name.

- Because, names are much easier to remember than Internet Protocol (IP) addresses,theattackerwillcreateatemporarymaliciouscloudtofaketheuseror customers.
- HenceusingIPaddressisnotalwaysfeasibleinDNSsincecustomerswillroute malicious cloud.
- It may happen that even after all the DNS security procedures are implemented, securityproblemswouldexistbasedonthemodeselectedbetweenthesenderand the receiver.

PhysicalSecurityIssues

- Physicalsecurity issuesaretheother partofthe cloud computing security.
- Althoughthedataisstoredinthevirtualserverinthecloud,itmustalsobestored in physical locations within physical hardware.
- Physicalsecurityinthecloudrepresentsthephysicalmachinesandstorageinthe datacenter.
- Physicalsecurityissuesshowsasalossofphysicalcontrol,humanattacks,power failure, access control, and third party trust.
- Thosephysicalissuesneedtobeprotectedalsofromanyinsiderandoutsider attackers.
- Usually the outside threats are easier to deal with than the inside threats because theoutsideattackshavebeenalreadypreparedforthroughriskassessmentplans.
- However,theinternalphysicalsecuritythreatsinclouddatacenterconstitutethe top risks in the cloud.

1) LossofPhysicalControl:

- The loss of physical access control occurs when the customers join the cloud eitherbykeepingtheirapplicationsinthecloud,orusingcloudstorageforsaving their data.
- This loss of control results in issues and concerns for the customers, such as trust andprivacyoftheirdatainthecloudprovider’sdatacenter,controlovertheirdata in the cloud, and legal restrictions by cloud provider

a) PrivacyandData

- Withprivate,publicandcommunityclouds,customer'sdatamaynotremainin thesamesystem.Inotherwords,itwillnotbelocatedinthecustomerpremises any more.

b) ControloverData

- Customersneedtohaveafullcontroloverthеirdata,andnotlimitedcontroland accountabilitywithinPubliccloudssuchas(IAAS)implementation,andthrough (PAAS) operations.
- Customersneedtohaveconfidencethattheproviderwillofferserviceswith appropriate controls.

c) LegalandRegulatoryCompliance

- Itmaybedifficultorunrealisticforcustomerstoutilizepubliccloudsiftheirdata need to be processed.
- Thisisasubjecttolegalrestrictionsorregulatorycompliance.
- Customersshouldexpectproviderstobuildandcertifytheircloudtoaddressthe needs of regulated markets, and achieve certifications and trust confidentiality between customers and providers of the services.

2) HumanAttacks:

- Human attacks happen when unauthorized personal tries to access the datacenter. Thisattackfordatacentercouldbemaninthemiddleattack,ormaliciousinsiders such as an employee of the datacenter.
- Thesekindsofattacksareexamplesofthecloudproviderlosingtheirsignificant control over securing datacenters and authorizing human attacker to enter their premises.

3) PowerFailure:

- In the event that the datacenterof cloud providers is faced with any kind of problems whichcausespowerfailure,andtheprovidersdonothaveanydisasterrecoveryplan, then the data in the cloud is at risk if it is not saved by the customer and user during the downtime.
- Thisgiverisetothepossibilityofattackersaccessingtheserversthroughthemanin the middle attacks .

- Amazon'scloudservicesinfrastructurefacedapowerfailureissueintheirdatacenter in August 2011 where many people who were using AWS were affected by such an outage because all the services were disconnect.